Chapter 4

Nuclear Reactions

4.1 INTRODUCTION

In Chapter 1, we have discussed how Rutherford discovered the existence of the nucleus by bombarding fast moving *a*-particles on thin gold foil. Similarly, Chadwick in 1932 bombarded fast moving *a*-particles on ^{9}Be nucleus, which led to the discovery of neutrons.

The question arises, what happens when we bombard fast moving, charged/neutral particles (also known as projectiles) on a stable nucleus (also called target nucleus)? Experimentally, it has been observed that these particles can interact with the nucleus or can be captured by the nucleus. The nucleus so formed can be different from the initial nucleus and is invariably unstable. It decays by emitting *g*-ray/a particle/a group of particles. The daughter nucleus so produced could be stable or unstable. In case it is unstable, it follows the laws of radioactive decay.

Nuclear Reaction

This process of bombarding a target nucleus by fast moving projectiles and the subsequent interaction between the two, which alters the composition, energy, etc. of the target nucleus is known as *nuclear reaction.* This is also known as transmutation of one element into another.

Suppose a target nucleus X is bombarded by particle a. During this process a new nucleus Y is formed and a particle (or *g*-ray) b is emitted, then this nuclear reaction is written as:

$$a + X \square Y + b$$

This reaction is also denoted as $X(a, b)Y$. For example, consider the following reaction:

$$^{4}_{2}\mathrm{He} + {}^{9}_{4}\mathrm{Be} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{1}_{0}n$$

It can be written as:

$$^{9}_{4}\mathrm{Be}\,(^{4}_{2}\mathrm{He},\ {}^{1}_{0}n)\,{}^{12}_{6}\mathrm{C}$$

or

$$^{9}_{4}\mathrm{Be}\,(\alpha,\ n)\,{}^{12}_{6}\,\mathrm{C}$$

If more than one particle say b_1, b_2 and b_3 are emitted, then the reaction is written as:

$$a + X \square\ Y + b_1 + b_2 + b_3$$

and it is denoted as $X(a, b_1\ b_2\ b_3)Y$. Consider the following reaction:

$$^{238}_{92}\mathrm{U} + {}^{14}_{7}\mathrm{N} \rightarrow {}^{243}_{97}\mathrm{Bk} + {}^{4}_{2}\mathrm{He} + {}^{1}_{0}n + {}^{1}_{0}n + {}^{1}_{0}n + {}^{1}_{0}n + {}^{1}_{0}n$$

Alternatively, we can write it as:

$$^{238}_{92}\mathrm{U} + {}^{14}_{7}\mathrm{N} \rightarrow {}^{243}_{97}\mathrm{Bk} + {}^{4}_{2}\mathrm{He} + 5\,{}^{1}_{0}n$$

The concise version is:

$$^{238}_{92}\mathrm{U}(^{14}_{7}\mathrm{N},\ \alpha\,5n)\,{}^{243}_{97}\mathrm{Bk}$$

In most of the nuclear reactions, mass and atomic number of both target and resulting nuclei are also written. Further, if the incident particle, emitted particles are composite particles (like heavy ions), their mass and charge numbers are also displayed. If there is no confusion, then in some cases, charge number is dropped.

The first nuclear reaction was performed by Rutherford in 1919. He bombarded nitrogen gas with *a*-particles obtained from natural radioactive source. The reaction is:

$$^{4}_{2}\mathrm{He} + {}^{14}_{7}\mathrm{N} \rightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{H}$$

or

$$^{14}_{7}\mathrm{N}(\alpha,\ p)\,{}^{17}_{8}\mathrm{O}$$

or

$$^{14}_{7}\mathrm{N}(^{4}_{2}\mathrm{He},\ {}^{1}_{1}\mathrm{H})\,{}^{17}_{8}\mathrm{O}$$

Similarly, the reaction used by Chadwick to discover neutron can be written as:

$$^{4}_{2}\mathrm{He} + ^{9}_{4}\mathrm{Be} \rightarrow ^{12}_{6}\mathrm{C} + ^{1}_{0}n$$

or

$$^{9}_{4}\mathrm{Be}(^{4}_{2}\mathrm{He}, ^{1}_{0}n)^{12}_{6}\mathrm{C}$$

It is not essential that nuclear reaction gets initiated only by particles emitted by radioactive materials. Nuclear reactions can be initiated by high energy charged particles. The charged particles can be imparted high energy by devices called *particle accelerators* like Cyclotron, Linear Accelerator, etc. (to be discussed in Chapter 6). In these devices charged particles are accelerated to very high energy and then they are bombarded on a target. These days accelerators are available that can accelerate particles starting from protons to uranium nuclei. The energy of *a*-particles emitted during radioactivity or by radioactive elements is limited up to 10 MeV. But with these devices, changed particles can be accelerated to very, very high energies, greater than few thousands of MeV.

The first nuclear reaction by accelerating protons was performed by Cockcroft and Walton in 1932.

$$^{7}_{3}\mathrm{Li} + ^{1}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + ^{4}_{2}\mathrm{He}$$

The study of nuclear reactions is extremely important. Almost all the properties of the nucleus such as mass, size, charge distribution, etc. have been obtained from such a study. Further, this study provides information about the nature of nuclear forces and reaction mechanisms.

Below we discuss some important types of nuclear reactions.

4.2 TYPES OF NUCLEAR REACTIONS

There are many ways of classifying nuclear reactions. Two commonly used ways to classify nuclear reactions are:

1. Reactions based on the reaction mechanism.
2. Reactions based on the mass of projectile.

4.2.1 Reactions Based on the Reaction Mechanism

Elastic Scattering

In elastic scattering incident particle strikes the target nucleus and leaves the nucleus without any energy loss or in elastic scattering there is no loss or gain of energy of the incident projectile. The direction of the incident particle generally changes. Some examples of elestic scattering are:

$$^{4}_{2}\mathrm{He} + {}^{208}_{82}\mathrm{Pb} \rightarrow {}^{208}_{82}\mathrm{Pb} + {}^{4}_{2}\mathrm{He}$$

$$^{1}_{0}n + {}^{208}_{92}\mathrm{U} \rightarrow {}^{208}_{92}\mathrm{U} + {}^{1}_{0}n$$

Inelastic Scattering

In inelastic scattering the incident particle loses some of its kinetic energy to the target nucleus. This raises the internal energy of the nucleus. The target nucleus is raised to the higher excited quantum state. Generally, this excited nucleus decays by *g*-emission. Some examples of inelastic scattering are:

$$^{1}_{1}\mathrm{H} + {}^{27}_{13}\mathrm{Al} \rightarrow {}^{27}_{13}\mathrm{Al}^{*} + {}^{1}_{1}\mathrm{H}$$

$$^{4}_{2}\mathrm{He} + {}^{56}_{26}\mathrm{Fe} \rightarrow {}^{56}_{26}\mathrm{Fe}^{*} + {}^{4}_{2}\mathrm{He}$$

where $*$ on the residual or product nucleus indicates that the nucleus is in excited state.

Radiative Capture

In such reactions incident particle is captured by the nucleus with the emission of *g*-rays. Some examples of radiative capture reaction are as under:

$$^{1}_{1}\mathrm{H} + {}^{27}_{13}\mathrm{Al} \rightarrow \left[{}^{28}_{14}\mathrm{Si}\right]^{*} \rightarrow {}^{28}_{14}\mathrm{Si} + \gamma$$

$$^{4}_{2}\mathrm{He} + {}^{12}_{6}\mathrm{C} \rightarrow \left[{}^{16}_{8}\mathrm{O}\right]^{*} \rightarrow {}^{16}_{8}\mathrm{O} + \gamma$$

here $*$ indicates that the nucleus is in excited state.

Compound Nuclear Reactions

The incident particle is absorbed by the target nucleus and a compound system is formed. This compound system lives for a long time ~10^{-14} to 10^{-15} seconds (compared to the time ~10^{-21} to 10^{-22} seconds taken by a low energy particle ~5 MeV to cross a distance equivalent to nuclear diameter)

and then a particle or group of particles is ejected. Some examples of compound nuclear reactions are:

$$^{4}_{2}\mathrm{He} + ^{27}_{13}\mathrm{Al} \rightarrow \left[^{31}_{15}\mathrm{P}\right] \rightarrow ^{30}_{15}\mathrm{P} + ^{1}_{0}n$$

$$^{4}_{2}\mathrm{He} + ^{60}_{28}\mathrm{Ni} \rightarrow \left[^{64}_{30}\mathrm{Zn}\right] \rightarrow ^{62}_{29}\mathrm{Cu} + ^{1}_{1}\mathrm{H} + ^{1}_{0}n$$

Photodisintegration

High energy *g*-ray photons are absorbed by the nucleus and the nucleus ejects a particle or a group of particles. For example,

$$^{2}_{1}\mathrm{H} + \gamma \rightarrow ^{1}_{1}\mathrm{H} + ^{1}_{0}n$$

$$^{16}_{8}\mathrm{O} + \gamma \rightarrow ^{11}_{6}\mathrm{C} + 2^{1}_{1}\mathrm{H} + 3^{1}_{0}n$$

Direct Reactions

In some cases fast moving incident particle while crossing through the nucleus or very close to the nucleus either picks up one or a few nucleons from the nucleus and keeps on moving in the forward direction or it loses one or a few nucleons to the nucleus and keep on moving in the forward direction.

First type of reactions are known as *pickup reactions* while the second type of reactions are known as *stripping reactions*. Both these reactions together are known as *direct reactions*. The typical reaction time of direct reactions is of the order of 10^{-21} to 10^{-22} seconds (compared with reaction time of compound reactions which is ~10^{-14} to 10^{-15} seconds). Some of the pickup reactions are:

$$^{59}_{27}\mathrm{Co} + ^{2}_{1}\mathrm{H} \rightarrow ^{57}_{26}\mathrm{Fe} + ^{4}_{2}\mathrm{He}$$

$$^{45}_{21}\mathrm{Sc} + ^{3}_{2}\mathrm{He} \rightarrow ^{44}_{21}\mathrm{Sc} + ^{4}_{2}\mathrm{He}$$

And some of the stripping reactions are:

$$^{63}_{29}\mathrm{Cu} + ^{2}_{1}\mathrm{H} \rightarrow ^{64}_{29}\mathrm{Cu} + ^{1}_{1}\mathrm{H}$$

$$^{197}_{79}\mathrm{Au} + ^{2}_{1}\mathrm{H} \rightarrow ^{198}_{80}\mathrm{Hg} + ^{1}_{0}n$$

Heavy Ion Reactions

The reaction in which an incident particle is a fast moving heavy ion is known as *heavy ion reaction*. For example, when $^{207}_{82}\mathrm{Pb}$ is accelerated to high

energy, i.e. $^{207}_{82}$Pb is a projectile and is bombarded on say $^{107}_{47}$Ag target

$$^{207}_{82}\text{Pb} + {}^{107}_{47}\text{Ag} \rightarrow {}^{106}_{47}\text{Ag} + {}^{208}_{82}\text{Pb}$$

Another such reaction is

$$^{27}_{13}\text{Al} + {}^{14}_{7}\text{N} \rightarrow {}^{16}_{8}\text{O} + {}^{25}_{12}\text{Mg}$$

Radioactive Decay

Typical examples of radioactive decay are *a*-, *b*- and *g*-decay. These have been discussed in detail in Chapter 3.

Fission Reactions

When heavy nuclei capture incident particle, they break into generally two fragments with the emission of few particles. The most common example is:

$$^{235}_{92}\text{U} + {}^{1}_{0}n \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3\,{}^{1}_{0}n$$

Fusion Reactions

When two light nuclei combine to form a heavier nucleus, the reaction is known as *fusion reaction*. For example,

$$^{2}_{1}\text{H} + {}^{2}_{1}\text{H} \rightarrow {}^{3}_{2}\text{He} + {}^{1}_{0}n$$
$$^{3}_{2}\text{He} + {}^{3}_{2}\text{He} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{1}\text{H} + {}^{1}_{1}\text{H}$$

Elementary Particle Production Reactions

When very high energy (E > 300 MeV) incident particles like protons, etc. interact with protons or neutrons, a variety of elementary particles like mesons, pions, etc. are produced. We shall discuss these reactions in detail in Chapter 8.

4.2.2 Reactions Based on the Mass of Projectile

Reactions Induced by Particles of Mass 1

Mass 1 particles are protons and neutrons. We discuss both the cases as follows:

(i) **Proton-induced reactions:**

(a) (*p*, *g*)*-reactions:* Radiative capture reactions have high probability for lighter elements and at low energy protons (E_p < 1 MeV).

General form of this reaction is:

$$\boxed{{}^{A}_{Z}X + {}^{1}_{1}\mathrm{H} \rightarrow {}^{A+1}_{Z+1}Y + \gamma}$$

Some specific cases are:

$${}^{27}_{13}\mathrm{Al} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{28}_{14}\mathrm{Si} + \gamma$$

$${}^{59}_{27}\mathrm{Co} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{60}_{28}\mathrm{Ni} + \gamma$$

(b) (*p*, *a*)-*reactions:* For majority of low *Z* nuclei, *Q*-value for (*p*, *a*)-reactions is +ve, i.e. these reactions are exothermic. These reactions predominate, if proton energy is between 2 MeV and 5 MeV. General form of this reaction is:

$$\boxed{{}^{A}_{Z}X + {}^{1}_{1}\mathrm{H} \rightarrow {}^{A-3}_{Z-1}Y + {}^{4}_{2}\mathrm{He}}$$

Some examples are:

$${}^{7}_{3}\mathrm{Li} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{4}_{2}\mathrm{He}$$

$${}^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{11}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He}$$

(c) (*p*, *n*)-*reactions:* The probability of (*p*, *n*)-reactions increases with increase in energy of protons. General form of this reaction is:

$$\boxed{{}^{A}_{Z}X + {}^{1}_{1}\mathrm{H} \rightarrow {}^{A}_{Z+1}Y + {}^{1}_{0}n}$$

Some typical cases are:

$${}^{23}_{11}\mathrm{Na} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{23}_{12}\mathrm{Mg} + {}^{1}_{0}n$$

$${}^{65}_{29}\mathrm{Cu} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{65}_{30}\mathrm{Zn} + {}^{1}_{0}n$$

At higher energies (*p*, 2*n*), (*p*, 3*n*), etc. reactions occur.

(ii) **Neutrons-induced reactions:** Neutrons have no electric charge and, therefore, can penetrate atomic nuclei without experiencing any Coulomb repulsion. So these reactions can occur at very low energies (~0.025 eV). At these low energies reactions with charged particles are not possible. Some of the neutron induced reactions are:

(a) (*n*, *g*)-*reactions:* These reactions have very high probability at energies ~0.025 eV (called thermal energy). In most of (*n*, *g*)-reactions, product nucleus formed is radioactive. The general form of these reactions is:

$$\boxed{{}^{A}_{Z}X + {}^{1}_{0}n \rightarrow {}^{A+1}_{Z}X + \gamma}$$

Some typical examples are:

$${}^{23}_{11}\text{Na} + {}^{1}_{0}n \rightarrow {}^{24}_{11}\text{Na} + \gamma \qquad (t_{1/2} \text{ of } {}^{24}_{11}\text{Na} = 15 \text{ hours})$$

$${}^{59}_{27}\text{Co} + {}^{1}_{0}n \rightarrow {}^{60}_{27}\text{Co} + \gamma \qquad (t_{1/2} \text{ of } {}^{60}_{27}\text{Co} = 5 \text{ years})$$

$${}^{24}_{12}\text{Mg} + {}^{1}_{0}n \rightarrow {}^{25}_{12}\text{Mg} + \gamma \qquad ({}^{25}_{12}\text{Mg stable})$$

(b) (n, p)-*reactions:* Since these reactions are endothermic (Q-value is –ve) so they occur at relatively high energies. These reactions are represented by the relation

$$\boxed{{}^{A}_{Z}X + {}^{1}_{0}n \rightarrow {}^{A}_{Z-1}Y + {}^{1}_{1}\text{H}}$$

Some examples are:

$${}^{27}_{13}\text{Al} + {}^{1}_{0}n \rightarrow {}^{27}_{12}\text{Mg} + {}^{1}_{1}\text{H}$$

$${}^{64}_{30}\text{Zn} + {}^{1}_{0}n \rightarrow {}^{64}_{29}\text{Cu} + {}^{1}_{1}\text{H}$$

(c) (n, a)-*reactions:* General form of these reactions is:

$$\boxed{{}^{A}_{Z}X + {}^{1}_{0}n \rightarrow {}^{A-3}_{Z-2}Y + {}^{4}_{2}\text{He}}$$

Some typical examples are:

$${}^{27}_{13}\text{Al} + {}^{1}_{0}n \rightarrow {}^{24}_{11}\text{Na} + {}^{4}_{2}\text{He}$$

$${}^{64}_{30}\text{Zn} + {}^{1}_{0}n \rightarrow {}^{61}_{28}\text{Ni} + {}^{4}_{2}\text{He}$$

(d) (n, $2n$)-*reactions:* These reactions have –ve Q-value, so high energy neutrons are needed for these reactions. These reactions are represented by

$$\boxed{{}^{A}_{Z}X + {}^{1}_{0}n \rightarrow {}^{A-1}_{Z}X + {}^{1}_{0}n + {}^{1}_{0}n}$$

Typical examples are:

$${}^{27}_{13}\text{Al} + {}^{1}_{0}n \rightarrow {}^{26}_{13}\text{Al} + {}^{1}_{0}n + {}^{1}_{0}n$$

$${}^{64}_{30}\text{Zn} + {}^{1}_{0}n \rightarrow {}^{63}_{30}\text{Zn} + {}^{1}_{0}n + {}^{1}_{0}n$$

Reactions Induced by Particles of Mass 2

Only projectile of mass 2 is deuteron.

(i) **Deuteron-induced reactions:** Deuteron is very loosely bound system (binding energy of deuteron ~2.2 MeV), so a large number of reactions are possible with deuterons and the most predominant being stripping and pickup reactions.

(a) (*d*, *p*)-*reactions:* The general (*d*, *p*)-reaction is:

$$\boxed{{}^{A}_{Z}X + {}^{2}_{1}\mathrm{H} \rightarrow {}^{A+1}_{Z}X + {}^{1}_{1}\mathrm{H}}$$

Examples are:

$${}^{27}_{13}\mathrm{Al} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{28}_{13}\mathrm{Al} + {}^{1}_{1}\mathrm{H}$$
$${}^{59}_{27}\mathrm{Co} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{60}_{27}\mathrm{Co} + {}^{1}_{1}\mathrm{H}$$

These reactions are also known as stripping reactions.

(b) (*d*, *n*)-*reactions:* The general form of this reaction is:

$$\boxed{{}^{A}_{Z}X + {}^{2}_{1}\mathrm{H} \rightarrow {}^{A+1}_{Z+1}Y + {}^{1}_{0}n}$$

Examples are:

$${}^{27}_{13}\mathrm{Al} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{28}_{14}\mathrm{Si} + {}^{1}_{0}n$$
$${}^{59}_{27}\mathrm{Co} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{60}_{28}\mathrm{Ni} + {}^{1}_{0}n$$

These are other examples of stripping reactions.

(c) (*d*, *t*)-*reactions:* These reactions are represented as:

$$\boxed{{}^{A}_{Z}X + {}^{2}_{1}\mathrm{H} \rightarrow {}^{A-1}_{Z}X + {}^{3}_{1}\mathrm{H}}$$

Typical examples are:

$${}^{27}_{13}\mathrm{Al} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{26}_{13}\mathrm{Al} + {}^{3}_{1}\mathrm{H}$$
$${}^{59}_{27}\mathrm{Co} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{58}_{27}\mathrm{Co} + {}^{3}_{1}\mathrm{H}$$

These reactions are also known as pick-up reactions. At high incident energies reactions such as (*d*, 2*n*), (*d*, 3*n*), (*d*, *pa*), etc. also become possible. However, the probability of (*d*, *g*)-reaction is very small.

Reactions Induced by Particles of Mass 3

In mass 3 category, we have 2 projectiles ${}^{3}_{1}\mathrm{H}$ and ${}^{3}_{2}\mathrm{He}$. Some of the reactions induced by ${}^{3}_{1}\mathrm{H}$ are:

$$^{59}_{27}\text{Co} + ^{3}_{1}\text{H} \rightarrow ^{60}_{27}\text{Co} + ^{2}_{1}\text{H} \qquad (t, d)\text{-reaction}$$

$$^{59}_{27}\text{Co} + ^{3}_{1}\text{H} \rightarrow ^{61}_{28}\text{Ni} + ^{1}_{0}n \qquad (t, n)\text{-reaction}$$

$$^{59}_{27}\text{Co} + ^{3}_{1}\text{H} \rightarrow ^{59}_{26}\text{Fe} + ^{3}_{2}\text{He} \qquad (t, ^{3}_{2}\text{He})\text{-reaction}$$

$$^{59}_{27}\text{Co} + ^{3}_{1}\text{H} \rightarrow ^{58}_{26}\text{Fe} + ^{4}_{2}\text{He} \qquad (t, ^{4}_{2}\text{He})\text{-reaction}$$

The reactions induced by $^{3}_{2}\text{He}$ are:

$$^{59}_{27}\text{Co} + ^{3}_{2}\text{He} \rightarrow ^{61}_{28}\text{Ni} + ^{1}_{1}\text{H} \qquad (^{3}_{2}\text{He}, p)\text{-reaction}$$

$$^{59}_{27}\text{Co} + ^{3}_{2}\text{He} \rightarrow ^{61}_{29}\text{Cu} + ^{1}_{0}n \qquad (^{3}_{2}\text{He}, n)\text{-reaction}$$

$$^{59}_{27}\text{Co} + ^{3}_{2}\text{He} \rightarrow ^{59}_{28}\text{Ni} + ^{3}_{1}\text{H} \qquad (^{3}_{2}\text{He}, t)\text{-reaction}$$

$$^{59}_{27}\text{Co} + ^{3}_{2}\text{He} \rightarrow ^{58}_{27}\text{Co} + ^{4}_{2}\text{He} \qquad (^{3}_{2}\text{He}, ^{4}_{2}\text{He})\text{-reaction}$$

$$^{59}_{27}\text{Co} + ^{3}_{2}\text{He} \rightarrow ^{60}_{27}\text{Co} + ^{1}_{1}\text{H} + ^{1}_{1}\text{H} \qquad (^{3}_{2}\text{He}, 2p)\text{-reaction}$$

Reactions Induced by Particles of Mass 4

The only projectile with mass 4 is *a*-particle. The various reactions are as under.

(a) (*a*, *p*)-*reactions:* The general form of the reaction is:

$$\boxed{^{A}_{Z}X + ^{4}_{2}\text{He} \rightarrow ^{A+3}_{Z+1}Y + ^{1}_{1}\text{H}}$$

Examples are:

$$^{27}_{13}\text{Al} + ^{4}_{2}\text{He} \rightarrow ^{30}_{14}\text{Si} + ^{1}_{1}\text{H}$$

$$^{59}_{27}\text{Co} + ^{4}_{2}\text{He} \rightarrow ^{62}_{28}\text{Ni} + ^{1}_{1}\text{H}$$

(b) (*a*, *n*)-*reactions:* The general form of (*a*, *n*)-reaction is:

$$\boxed{^{A}_{Z}X + ^{4}_{2}\text{He} \rightarrow ^{A+3}_{Z+2}Y + ^{1}_{0}n}$$

Examples are:

$$^{27}_{13}\text{Al} + ^{4}_{2}\text{He} \rightarrow ^{30}_{15}\text{P} + ^{1}_{0}n$$

$$^{59}_{27}\text{Co} + ^{4}_{2}\text{He} \rightarrow ^{62}_{29}\text{Cu} + ^{1}_{0}n$$

(c) (*a,* 2*n*)-*reactions:* The general form of this reaction is:

$$\boxed{^{A}_{Z}X + ^{4}_{2}\text{He} \rightarrow ^{A+2}_{Z+2}Y + 2^{1}_{0}n}$$

Typical examples are:

$$^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{29}_{15}\mathrm{P} + 2\,{}^{1}_{0}n$$

$$^{59}_{27}\mathrm{Co} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{61}_{29}\mathrm{Cu} + 2\,{}^{1}_{0}n$$

At higher energies reactions such as (a, $3n$), (a, $4n$), (a, $5n$), etc. have been observed.

Reactions Induced by Heavy Ions

Heavy ions are generally projectiles with mass greater than a-particles. Depending upon the energy of heavy ions large variety of reactions are possible. Some of the heavy ion reactions are:

$$^{27}_{13}\mathrm{Al}({}^{14}_{7}\mathrm{N},\ {}^{16}_{7}\mathrm{N})\,{}^{25}_{13}\mathrm{Al}$$

$$^{27}_{13}\mathrm{Al}({}^{14}_{7}\mathrm{N},\ {}^{14}_{8}\mathrm{O})\,{}^{27}_{12}\mathrm{Mg}$$

$$^{27}_{13}\mathrm{Al}({}^{12}_{6}\mathrm{C},\ \alpha\,2n)\,{}^{33}_{17}\mathrm{Cl}$$

$$^{197}_{79}\mathrm{Au}({}^{12}_{6}\mathrm{C},\ 4n)\,{}^{205}_{85}\mathrm{At}$$

$$^{238}_{92}\mathrm{U}({}^{14}_{7}\mathrm{N},\ \alpha\,5n)\,{}^{243}_{97}\mathrm{Bk}$$

Generally, heavy ion reactions have been used to synthesize elements with atomic number > 92.

Reactions Induced by g-rays

The first reactions induced by g-rays are:

$$^{2}_{1}\mathrm{H} + \gamma \rightarrow {}^{1}_{1}\mathrm{H} + {}^{1}_{0}n$$

$$^{9}_{4}\mathrm{Be} + \gamma \rightarrow {}^{8}_{4}\mathrm{Be} + {}^{1}_{0}n$$

Gamma rays used for these reactions were 2.614 MeV from ^{208}Tl. When g-rays of very high energy are available, number of reactions induced by g-rays are observed. Some of these reactions are:

$$^{31}_{15}\mathrm{P} + \gamma \rightarrow {}^{30}_{15}\mathrm{P} + {}^{1}_{0}n$$

$$^{10}_{5}\mathrm{B} + \gamma \rightarrow {}^{8}_{4}\mathrm{Be} + {}^{2}_{1}\mathrm{H}$$

$$^{11}_{5}\mathrm{B} + \gamma \rightarrow {}^{8}_{4}\mathrm{Be} + {}^{3}_{1}\mathrm{H}$$

These reactions induced by *g*-rays are also known as *photodisintegrations* or *photonuclear reactions*.

Nuclear Decay and Nuclear Reactions

Dominant differences between nuclear decay (Chapter 3) and nuclear reactions are shown in Table 4.1

TABLE 4.1 Difference between nuclear decay and nuclear reactions

Nuclear decay	Nuclear reaction
It is a spontaneous process (i.e. no external aid is needed)	It is not a spontaneous process (i.e. external aid is needed to start a reaction)
Only *a*-, *b*- and *g*-emissions are possible	In nuclear reactions all possible particles like *p*, *n*, ^{2}H, ^{3}He, ^{4}He, ^{6}Li, etc. can be emitted
Mass of the daughter nucleus either remains same or decreases	Mass of the product nucleus can also increase
Energy of the emitted particles or *g*-rays is generally low	Energy of the emitted particles or *g*-rays can be very high

4.3 NUCLEAR REACTION CROSS-SECTION

A question often arises that when some projectiles are bombarded on a target, an interaction or reaction between these two takes place, then how many nuclei are taking part in this reaction? Is this number small or large? Depending upon the numbers, the probability of a particular reaction becomes small or large. There must be a quantitative measure of this probability. This quantity should be easily measurable and also theoretically calculable, so that we can compare the experimental results with the theoretical calculations.

The quantity, which defines the probability of a given reaction, is called the *nuclear reaction cross-section* and it is denoted as *s*.

The reaction cross-section can be visualized as an effective area around the nucleus such that if the incident particles cross that area, a nuclear reaction takes place, otherwise not. This area is also known as effective area. It is not exactly same as the geometrical area (pr^2), where *r* is the radius of the nucleus. This effective area could be larger, smaller or equal to the geometrical area of the nucleus. Thus, nuclear cross-section is a measure of the probability that bombarding particles at a particular energy would interact with the target.

4.3.1 Measurement of Cross-Section

Consider a beam of particles of flux I (number of particles passing a unit area per unit time) incident on a thin sheet of a material having thickness dx and face area A as shown in Figure 4.1. Further assume that thickness is so small that none of the nuclei of the thin sheet overlap each other. Thus, each nucleus has an equal probability to cause a nuclear reaction with the incident beam. Let s be the effective area of the target nucleus. As stated earlier, this area is such that when the incident particle crosses this area the reaction always takes place. Let n be the number of target nuclei per cc in this thin sheet.

Number of nuclei in the sheet per unit face area = $n \square dx$

Total number of nuclei in the sheet = $A \square n \square dx$

Since each nucleus has an effective area s, so the total effective area available for the reaction = $A \square n \square s \square dx$

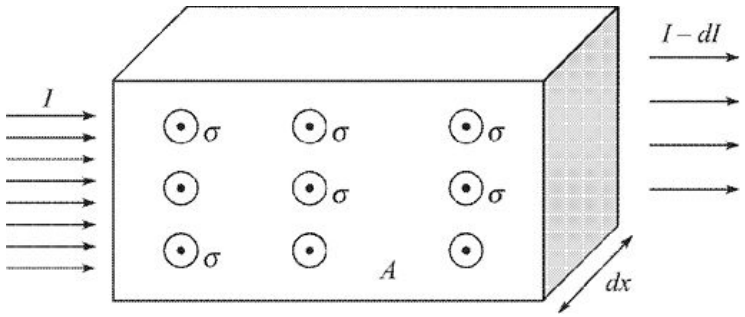


Figure 4.1 A beam of particles incident on a thin foil of face area A and thickness dx.

$$\text{Fractional effective area } f = \frac{\text{Total effective area}}{\text{Total face area}} = \frac{\sigma \times A \times n \times dx}{A}$$

$$= \sigma \times n \times dx$$

When a projectile takes part in a nuclear reaction, it is absorbed by the target nucleus, so the flux of incident particles decreases. If the fractional effective area (f) is large, then the corresponding decrease in the flux (I) is large and vice versa.

Thus, the change in the intensity dI is given by

$$dI = -fI$$

substituting for f

$$dI = -Isndx$$

or

$$\frac{dI}{I} = -\sigma n dx$$

–ve sign has been introduced because as dx increases the flux I decreases.

Assuming that at $x = 0$, $I = I_0$, integrating equation with respect to x, we get

Thus, the flux of the incident beam exponentially decreases as the thickness of the sheet increases.

$$\boxed{I = I_0 e^{-n\sigma x}}$$

4.3.2 Units of Cross-Section

Units of reaction cross-sections are the units of area. These cross-sections are normally very small ~10^{-27} to 10^{-28} m^2. Since this value is small so it is convenient to use a unit called *barn*.

$$\boxed{1 \text{ barn (b)} = 10^{-28} \text{ m}^2}$$

Smaller units of cross-sections are millibarn (mb), microbarn (μb), etc.

$$\boxed{1 \text{ mb} = 10^{-3} \text{ barn} = 10^{-31} \text{ m}^2}$$

$$\boxed{1 \mu\text{b} = 10^{-6} \text{ barn} = 10^{-34} \text{ m}^2}$$

4.3.3 Different Types of Cross-Sections

In nuclear reactions we come across different types of reaction cross-sections. They are:

1. Partial reaction cross-section.
2. Total reaction cross-section.
3. Differential reaction cross-section.

Partial Reaction Cross-Section

For a given energy and target projectile combination, several nuclear reactions such as elastic scattering, inelastic scattering, capture reaction, disintegration reactions, etc. are possible simultaneously. There is a definite reaction cross-section for each such reaction denoted by s_1, s_2, s_3 ,…, etc. These cross-sections are known as *partial reaction cross-sections*.

Total Reaction Cross-Section

As stated earlier, for a given energy and target projectile combination, many partial cross-sections may exist simultaneously. Sum of all such partial cross-sections is known as *total reaction cross-section s*.

$$s = s_1 + s_2 + s_3 + \ldots$$

Differential Reaction Cross-Section

When the incoming particles interact with the target nuclei nuclear reaction or scattering takes place. Very often the outgoing particles have an anisotropic distribution. This means that the number of outgoing particles at different angles will be different. Also in most of the cases, the energy of the outgoing particles may be different at different angles. We measure the number of outgoing particles per second into a solid angle $d\square$ making an angle q with the incident direction. For measuring this quantity, another type of cross-section is introduced. This cross-section depends upon the angle and is called *differential cross-section*. This cross-section is defined as the cross-section per unit solid angle and is denoted by $s(q, f)$.

$$\sigma(\theta, \phi) = \frac{d\sigma}{d\Omega}$$

And the total cross-section is given by

$$\sigma_T = \int_\Omega \frac{d\sigma}{d\Omega} d\Omega$$

4.4 CONSERVATION LAWS IN NUCLEAR REACTIONS

In nuclear reactions certain physical quantities never change during and after the reaction. We say that these quantities are always conserved in nuclear reactions. Some of the conservation laws are as under.

4.4.1 Conservation of Mass-Energy

In a nuclear reaction total energy of reactants (rest mass plus kinetic energy for a non-relativistic case) is always equal to the total energy of the products (rest mass plus kinetic energy). Consider a reaction

$$a + X \square Y + b$$

If K_a, K_b and K_Y are kinetic energies of a, b and Y respectively, then

$$K_a + m_a c^2 + m_X c^2 = K_Y + m_Y c^2 + K_b + m_b c^2$$

assuming target nucleus X is at rest initially. However, for relativistic case, we have

$$E_a + m_X c^2 = E_Y + E_b$$

where

$$E_i = \sqrt{p_i^2 c^2 + m_i^2 c^4}$$

where $i = a, y$ or b

4.4.2 Conservation of Linear Momentum

In a nuclear reaction total initial momentum must be equal to the total final momentum. Assuming the nucleus of the target is at rest, the momentum brought into the nucleus by the projectile is distributed among the products.

4.4.3 Conservation of Charge/Atomic Number

In a nuclear reaction total charge on the reactants must be equal to the total charge on products, i.e. nuclear charge cannot be created or destroyed.

$$^{27}_{13}\mathrm{Al} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{24}_{11}\mathrm{Na} + 3{}^{1}_{1}\mathrm{H} + {}^{1}_{0}n$$

Here, total charge on reactants is 13 + 1 = 14. Total charge on products is 11 + 3 □ 1 = 14. Similarly,

$${}^{1}_{0}n \rightarrow {}^{1}_{1}\mathrm{H} + {}^{0}_{-1}e + \bar{\nu}$$

Total charge on the reactants is 0. Total charge on products is 1 + (–1) = 0.

4.4.4 Conservation of Nucleons/Mass Number

In a nuclear reaction total number of nucleons before and after the reaction remains the same, i.e. nucleons cannot be created nor destroyed in a nuclear reaction. For example,

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3{}^{1}_{0}n$$

There are 235 + 1 = 236 nucleons before the reaction and 141 + 92 + 3 □ 1 = 236 nucleons after the reaction.

4.4.5 Conservation of Angular Momentum

The total angular momentum I which is the vector sum of spin angular momentum s and relative orbital angular momentum l of the two sides must be equal.

4.4.6 Conservation of Spin

This law states that if the vectorial sum of the spins of the reactants is an odd integer multiple of □/2, then the vectorial sum of the spin of products must also be an odd integer multiple of □/2. Consider the decay of neutron

$$^{1}_{0}n \rightarrow {}^{1}_{1}\mathrm{H} + {}^{0}_{-1}e + \bar{\nu}$$

Since the spin of all the particles involved in the reaction is 1/2, so the left-hand side has spin
1/2 and the right-hand side can have spin 1/2 or 3/2. Thus, this satisfies the law of conservation of spin.
Similarly, if the vectorial sum of the spins of the reactants is an integer multiple of □, then the vectorial sum of the spin of products must also be an integer multiple of □.

4.4.7 Conservation of Statistics

Spin ½ particles follow Fermi–Dirac statistics, while integer spin particles follow Bose–Einstein statistics. If the number of particles involved as reactants is odd, then the number of product particles must be odd, so the reaction follows the Fermi–Dirac statistics.

Similarly, if the number of particles involved as reactants is even, then the number of product particles must be even, so the reaction follows the Bose–Einstein statistics. This behaviour is consistent with conservation of nucleon number as discussed before. For example, consider the reaction

$$^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{30}_{15}\mathrm{P} + {}^{1}_{0}n$$

Total mass number (A) on right-hand side and left-hand side is 31. So, both sides follow the Fermi–Dirac statistics.

4.4.8 Conservation of Parity

Total parity in a nuclear reaction must be conserved, i.e. the total parity of the reactants must be equal to the total parity of the products. No violation of parity conservation has been reported in nuclear reactions based on strong interactions.

4.4.9 Conservation of Lepton Number

Lepton number of the reactants must be equal to that of products. Leptons like e^-, n, m^-
etc. are assigned total lepton number as +1, while their antiparticles have been assigned total lepton number –1. For the reaction

$$^1_1\mathrm{H} \rightarrow {}^1_0n + e^+ + \nu$$

For left-hand side of the equation, there are no leptons, so the lepton number is 0. On right-hand side, positron has lepton number –1, while for neutrino, lepton number is +1. So the total lepton number on right-hand side is –1 + 1 = 0 (see also Section 8.7.2).

4.5 KINEMATICS OF NUCLEAR REACTIONS

In any nuclear reaction, laws of conservation of energy and momentum always hold good. These conservation laws impose certain conditions on nuclear reactions like that the energy of the outgoing particles must be real, etc. These restrictions are called kinematical restrictions and the mathematical relations derived by imposing these restrictions are known as *kinematics*.
Consider the following nuclear reaction in which a light projectile of mass M_1 of kinetic energy E_1 interacts with a nucleus of mass M_2 (assumed stationary), the outgoing light particle is of mass M_3 of energy E_3 and the product heavy nucleus is of mass M_4 of energy E_4.

$$M_1 + M_2 \square M_3 + M_4$$

Applying the law of conservation of mass-energy assuming that the nucleus of mass M_2 initially is at rest, then

$$M_1c^2 + E_1 + M_2c^2 = E_3 + M_3c^2 + E_4 + M_4c^2$$

$$E_3 + E_4 - E_1 = [(M_1 + M_2) - (M_3 + M_4)]c^2 \qquad (4.1)$$

Define a quantity Q as the difference between the kinetic energies of products and that of the incident particle in a reaction. Therefore,

$$Q = E_3 + E_4 - E_1$$

Using this value of Q, Eq. (4.1) becomes

$$Q = (M_1 + M_2)c^2 - (M_3 + M_4)c^2 \qquad (4.2)$$

Thus, Q-value of the reaction is the difference between masses of the interacting particles to that of the products.

If Q is positive, i.e. total mass of the reactants is more than the total mass of the products, then such reactions are known as *Exoergic* or *Exothermic reactions*. The second possibility is that Q is negative, i.e. mass of the reactants is less than that of products, such reactions are known as *Endoergic* or *Endothermic reactions*.

Schematically a nuclear reaction is represented in Figure 4.2, where a particle of mass M_1, kinetic energy E_1 and momentum $\sqrt{2M_1E_1}$ strikes a stationary target of mass M_2 resulting in the formation of a compound nucleus of mass M_c. This compound nucleus decays into lighter particle of mass M_3, kinetic energy E_3 and momentum $\sqrt{2M_3E_3}$ and the heavy nucleus of mass M_4, kinetic energy E_4 and momentum $\sqrt{2M_4E_4}$.

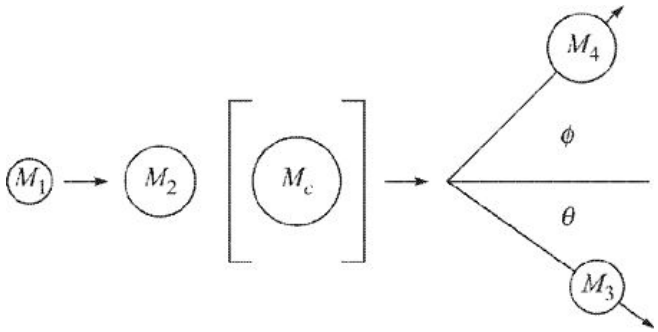


Figure 4.2 Dynamics of two-body nuclear reaction in laboratory frame.

Applying law of conservation of energy to this reaction, we get

$$E_1 + Q = E_3 + E_4 \qquad (4.3)$$

Using law of conservation of linear momentum in a direction of incident particle and perpendicular to it, we get the following equations:

$$\sqrt{2M_1E_1} = \sqrt{2M_3E_3}\cos\theta + \sqrt{2M_4E_4}\cos\phi \tag{4.4}$$

$$0 = \sqrt{2M_3E_3}\sin\theta - \sqrt{2M_4E_4}\sin\phi \tag{4.5}$$

Equations (4.2) to (4.5) are valid only if the system of masses M_1, M_2, M_3 and M_4 is an isolated system. (By isolated system we mean that none of the particles M_1, M_2, M_3 and M_4 are interacting with any other particle present in the system.) The most common practice is to perform experiments on ejected light particle of mass M_3. Therefore, using Eqs. (4.3) to (4.5), we eliminate E_4 and angle f.

Equations (4.4) and (4.5) can be written as:

$$\sqrt{2M_4E_4}\cos(\phi) = \sqrt{2M_1E_1} - \sqrt{2M_3E_3}\cos\theta$$

$$\sqrt{2M_4E_4}\sin(\phi) = \sqrt{2M_3E_3}\sin\theta$$

Squaring these two equations and adding, we get

$$2M_4E_4 = 2M_1E_1 + 2M_3E_3 - 4\sqrt{M_1M_3E_1E_3}\cos\theta$$

or

$$E_4 = \frac{M_1}{M_4}E_1 + \frac{M_3}{M_4}E_3 - \frac{2}{M_4}\sqrt{M_1M_3E_1E_3}\cos\theta$$

Substituting E_4 in Eq. (4.3), we get

$$E_1 + Q = E_3 + \frac{M_1}{M_4}E_1 + \frac{M_3}{M_4}E_3 - \frac{2}{M_4}\sqrt{M_1M_3E_1E_3}\cos\theta$$

or

$$Q = \frac{M_1}{M_4}E_1 - E_1 + \frac{M_3}{M_4}E_3 + E_3 - \frac{2}{M_4}\sqrt{M_1M_3E_1E_3}\cos\theta \tag{4.6}$$

$$\boxed{Q = \frac{M_1 - M_4}{M_4}E_1 + \frac{M_3 + M_4}{M_4}E_3 - \frac{2}{M_4}\sqrt{M_1M_3E_1E_3}\cos\theta} \tag{4.7}$$

This equation is known as *Q-value equation*. Here kinetic energy of the incident particle E_1 and kinetic energy of the outgoing light particle E_3 can accurately be measured experimentally.

In many applications, actual masses of interacting particles can be replaced by mass number A_1, A_2, A_3 and A_4 without any significant error.

The Q-value equation (4.7) is quadratic in $\sqrt{E_3}$. So, the solution of this quadratic equation is:

$$\sqrt{E_3} = v \pm \sqrt{v^2 + w} \tag{4.8}$$

where

$$v = \frac{\sqrt{M_1 M_3 E_1}}{M_3 + M_4} \cos\theta \tag{4.9}$$

$$w = \frac{M_4 Q + E_1(M_4 - M_1)}{M_3 + M_4} \tag{4.10}$$

According to this equation only those reactions are energetically possible for which $\sqrt{E_3}$ is real and positive.

4.5.1 Exoergic or Exothermic Reactions

In such reactions the total mass of the reactants is more than that of the products. This excess mass appears as the kinetic energy of the products. Let us study E_3 and direction q of the particle M_3 as a function of incident particle energy E_1, which is gradually increased.

Zero Bombarding Energy ($E_1 \cong 0$)

If $E_1 \square 0$, then from Eq. (4.9)

$$v \to 0$$

and from Eq. (4.10)

$$w = \frac{QM_4}{M_3 + M_4}$$

From Eq. (4.8)

$$E_3 = \frac{QM_4}{M_3 + M_4}$$

Therefore, the kinetic energy E_3 of the outgoing particle is independent of angle q.

Finite Bombarding Energy ($E_1 > 0$)

Generally, the projectile of mass M_1 is lighter than the unobserved nucleus of mass M_4. Therefore, the term $M_4 - M_1$ is always positive or from Eq. (4.10),

w is always positive for all values of E_1. Then from Eq. (4.8) only one value of E_3 is positive. We call this situation as E_3 is single-valued for all bombarding energies and is given by Eq. (4.8). Here E_3 depends upon angle q and would be smallest at $q = 180°$.

4.5.2 Endoergic or Endothermic Reactions

When E_1 □ 0 or zero bombarding energy and Q is negative, then from Eqs. (4.9) and (4.10),

v □ 0 and w is negative. Therefore, $v^2 + w$ is always negative. Hence, $\sqrt{E_3}$ is imaginary. This means no endothermic reaction is possible.

4.5.3 Threshold Energy

The smallest bombarding energy at which nuclear reaction just starts is known as threshold energy, $(E_1)_{th}$. According to Eq. (4.8) reaction will take place only if E_1 is large enough to make $v^2 + w = 0$. From Eqs. (4.9) and (4.10)

$$\frac{M_1 E_1 M_3}{(M_3 + M_4)^2}\cos^2\theta + \frac{M_4 Q + E_1(M_4 - M_1)}{M_3 + M_4} = 0$$

Since $M_3 + M_4$ □ 0, so multiplying by $M_3 + M_4$ on both sides, we have

$$\frac{M_1 E_1 M_3}{(M_3 + M_4)}\cos^2\theta + M_4 Q + E_1(M_4 - M_1) = 0$$

Substituting for $\cos^2(\theta) = 1 - \sin^2\theta$, we get

$$\frac{M_1 E_1 M_3}{(M_3 + M_4)}[1 - \sin^2\theta] + E_1(M_4 - M_1) = -M_4 Q$$

Taking E_1 as common and rearranging, we have

$$E_1\left[\frac{M_1M_3}{M_3+M_4}+M_4-M_1-\frac{M_1M_3}{M_3+M_4}\sin^2\theta\right]=-M_4Q$$

Multiplying by $M_3 + M_4$ on both sides

$$E_1\left[M_3M_4+M_4^2-M_1M_4-M_1M_3\sin^2\theta\right]=-M_4Q(M_3+M_4)$$

or

$$E_1=\frac{-M_4Q(M_3+M_4)}{M_3M_4+M_4^2-M_1M_4-M_1M_3\sin^2\theta}$$

Dividing the numerator and denominator by M_4, E_1 at an angle q is given by

$$(E_1)_\theta=-Q\frac{M_3+M_4}{M_3+M_4-M_1-\dfrac{M_1M_3}{M_4}\sin^2\theta} \tag{4.11}$$

If light outgoing particle M_3 is observed at an angle $q = 0°$, then $(E_1)q$ has its minimum possible value, known as *threshold energy* and is given by

$$(E_1)_{\text{th}}=-Q\frac{M_3+M_4}{M_3+M_4-M_1} \tag{4.12}$$

Equation (4.2) can be written as:

$$M_1+M_2=M_3+M_4+\frac{Q}{c^2}$$

Generally,

$$M_2 >> \frac{Q}{c^2}$$

Therefore, neglecting $\frac{Q}{c^2}$, we have

$$M_1+M_2=M_3+M_4$$

Using this result, Eq. (4.12) becomes

$$\boxed{(E_1)_{\text{th}}=-Q\frac{M_1+M_2}{M_2}}$$

At the threshold energy, outgoing particle with smaller mass (M_3) appears at $q = 0°$. When the bombarding energy is increased M_3 appears at larger angles i.e. $q > 0°$ also.

4.6 COMPOUND NUCLEUS

In 1936 Bohr proposed the compound nucleus hypothesis. According to this hypothesis, nuclear reaction takes place in two stages:

1. Formation of compound nucleus.
2. Decay of compound nucleus.

Formation of Compound Nucleus

The incident particle strongly interacts with the nucleons present in the nucleus, giving away its energy to all the nucleons, thereby losing its identity and becomes a part of the nucleus. This new nucleus formed is in a highly excited state and is called compound nucleus.

Decay of Compound Nucleus

There is a finite probability that one or a group of nucleons residing on the surface or near the surface of the nucleus and having enough energy can escape the compound nucleus, thereby leading to the decay of compound nucleus. This reaction can be written as

$$M_1 + M_2 \square [M_c] \square M_3 + M_4$$

This hypothesis also implies that the decay of compound nucleus is independent of the way it was formed. This independence of mode of formation and decay is true only if the decay time of the compound nucleus is much longer than the nuclear time. Nuclear time is defined as the time taken by an incident particle to cross the diameter of the nucleus. For example, 5 MeV proton has velocity ~$3 \square 10^7$ m/s. Nuclear diameter of a nucleus with mass 200 is about ~$7 \square 10^{-15}$ m. Thus, nuclear time is ~$7 \square 10^{-15}/3 \square 10^7 = 2 \square 10^{-22}$ s. Decay time of the compound nucleus has been experimentally measured and is about 10^{-14} to 10^{-15} s. This is about 10^7 times larger than the nuclear time.

The compound nucleus forgets the mode of formation. However, decay of the compound nucleus depends upon the energy, angular momentum and parity brought in by the incident particle.

Evidence to the compound nucleus hypothesis was provided by the series of experiments of Ghoshal. In his experiments compound nucleus $\left[{}^{64}_{30}\mathrm{Zn}\right]^*$ was formed in two different modes as under:

$$ {}^{1}_{1}\mathrm{H} + {}^{63}_{29}\mathrm{Cu} \rightarrow \left[{}^{64}_{30}\mathrm{Zn}\right]^* $$

and

$$ {}^{4}_{2}\mathrm{He} + {}^{60}_{28}\mathrm{Ni} \rightarrow \left[{}^{64}_{30}\mathrm{Zn}\right]^* $$

This compound nucleus decays by one of the following modes:

$$ \left[{}^{64}_{30}\mathrm{Zn}\right]^* \rightarrow {}^{63}_{30}\mathrm{Zn} + {}^{1}_{0}n $$

$$ \rightarrow {}^{62}_{30}\mathrm{Zn} + 2{}^{1}_{0}n $$

$$ \rightarrow {}^{62}_{29}\mathrm{Cu} + {}^{1}_{0}n + {}^{1}_{1}\mathrm{H} $$

Ghoshal found that the decay probability of the compound nucleus was independent of the mode of its formation.

4.7 NUCLEAR FISSION

Nuclear fission is a process of splitting a nucleus, generally heavy atom (target) into two or more lighter atoms (fission fragments) when it is bombarded by neutrons or charged particles. A few radioactive nuclides can also spontaneously fission such as ^{254}Cf and ^{256}Fm. Fission releases large amount of energy along with one or more neutrons. The large amount of energy is released as sum of masses of the fission products is less than the original mass of the heavy atom and bombarding particle. This missing mass (□0.1% of the heavy atom mass) is converted into energy according to Einstein's equation $E = mc^2$.

4.7.1 Neutron-Induced Fission

Fermi and his associates bombarded natural uranium with slow neutrons and found at least four b^--activities having different half-lives. They assumed that these activities could be the product of an (n, g) reaction leading to the formation of elements with atomic numbers 93, 94, etc. through successive b^--decay as shown under.

$$^{238}_{92}\mathrm{U} + n \rightarrow {}^{239}_{92}\mathrm{U}$$

$$^{239}_{92}\mathrm{U} \rightarrow {}^{239}_{93}\mathrm{Np} + \beta^-$$

$$^{239}_{93}\mathrm{Np} \rightarrow {}^{239}_{94}\mathrm{Pu} + \beta^-$$

However, after extremely painstaking experiments, Hahn and Strassmann in 1938 established that isotopes of Ba, La, Sr, Y, Rb, Cs, etc. were among the sources of b^--activity. They concluded that these radioactive nuclides were formed as a consequence of the uranium nucleus splitting into two (not necessarily equal) parts following the bombardment of slow neutrons on uranium nucleus. This is known as *neutron-induced fission*. (Another type of fission is spontaneous fission, in which a heavy nucleus with excess of neutrons fissions into two or more smaller fragments. This process is similar to neutron-induced fission, with the exception that no neutrons are needed to initiate the fission. Examples of such fissionable nuclei are ^{254}Cf, ^{256}Fm, etc.)

Meitner and Frisch theoretically predicted that during this splitting process, □200 MeV energy is released and they named this process as *Nuclear Fission*. This prediction was experimentally verified by Frisch. In 1939 Bohr and Wheeler explained the phenomenon of fission on the basis of liquid drop model of the nucleus. They also established that with slow neutrons only $^{235}_{92}$U undergo fission. The fission of $^{235}_{92}$U with slow neutrons is diagrammatically shown in Figure 4.3. Later on it was established that fission can be induced in almost all the nuclei through interaction of neutrons, protons, *a*-particles or even heavy nuclei of suitable kinetic energy.

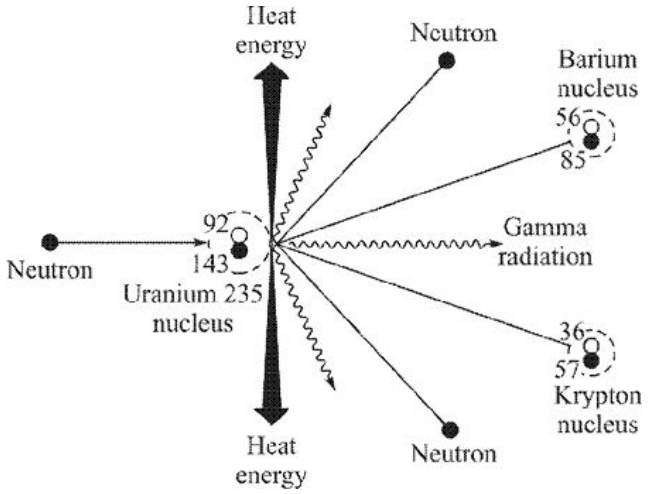


Figure 4.3 Schematic diagram of fission of ^{235}U by neutrons.

Some of the fission reactions are

$$^{235}_{92}\text{U} + {}^{1}_{0}n \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3{}^{1}_{0}n \quad Q = +175 \text{ MeV}$$

$$^{63}_{29}\text{Cu} + p \rightarrow {}^{24}_{11}\text{Na} + {}^{39}_{19}\text{K} + {}^{1}_{0}n \quad Q = -24 \text{ MeV}$$

$$^{2}_{1}\text{H} + \gamma \rightarrow {}^{1}_{1}\text{H} + {}^{1}_{0}n \quad Q = -2.2 \text{ MeV}$$

Out of all the possible fission reactions, fission induced by slow or thermal neutrons (having energy 0.025 eV) is most commonly studied fission reaction. This reaction is utilized in nuclear reactors to generate electricity and to produce radioactive isotopes. It has been found that along with $^{235}_{92}$U, other heavy nuclei which can undergo fission with thermal neutrons are $^{233}_{92}$U and $^{239}_{94}$Pu.

When nuclei undergo fission, the fission products are not determined uniquely. There is a distribution of masses of two fission products as shown in Figure 4.4. The distribution is symmetric. About 97% of the total fission products fall within the narrow range from mass 85 to 104 (lighter fragment group) and the other range from mass 130 to 149 (heavier group). For heavy fragment, there is always a corresponding light fragment. The maximum yield of fragments is at about A_1 = 95 and A_2 = 140. In all more than 200 fission fragments have been observed.

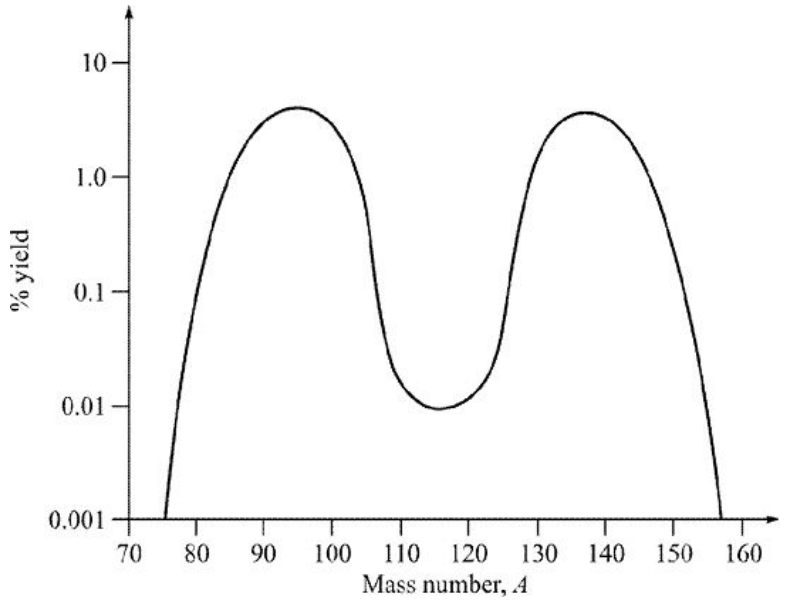


Figure 4.4 Per cent yield of different fission fragments in fission of ^{235}U.

4.7.2 Energy Released in Fission

When heavy nuclei undergo fission, large amount of energy is released. It can be shown by considering the fission of $^{235}_{92}U$ by thermal neutrons. The fission fragments formed have mass numbers about 95 and 139. These fission fragments are radioactive and undergo several stages of b^--decay. Ultimately after emission of seven b^--particles, stable nuclides ^{95}Mo and ^{139}La are formed. This event is finally represented by the following reaction:

$$^{235}_{92}\text{U} + ^{1}_{0}n \rightarrow ^{95}_{42}\text{Mo} + ^{139}_{57}\text{La} + 7\,^{0}_{-1}e + 2\,^{1}_{0}n$$

Let us consider the masses of reactants and products in this reaction.

Masses of reactants:

$$\text{Mass of } ^{235}_{92}\text{U} = 235.0439 \text{ amu}$$

$$\text{Mass of } ^{1}_{0}n = 1.0087 \text{ amu}$$

$$\text{Total mass} = 236.0526 \text{ amu}$$

Masses of products:

$$\text{Mass of } ^{139}_{57}\text{La} = 138.9061 \text{ amu}$$

$$\text{Mass of } ^{95}_{42}\text{Mo} = 94.9058 \text{ amu}$$

$$\text{Mass of } 7\,^{0}_{-1}e = 7 \times 0.00055 = 0.00385 \text{ amu}$$

$$\text{Mass of } 2\,^{1}_{0}n = 2 \times 1.0087 = 2.0174 \text{ amu}$$

$$\text{Total mass} = 235.8332 \text{ amu}$$

It is evident from the above calculations that the products are lighter than reactants by
0.219 amu, which is converted into energy, i.e. 0.219 □ 931.481 □ 204 MeV. In order to get an idea about the energy evolved in fission, let us calculate the energy evolved when 1 g of $^{235}_{92}U$ undergoes fission.

Since 1 MeV = 1.6×10^{-6} erg, therefore 204 MeV = $204 \times 1.6 \times 10^{-6}$ erg = 3.2×10^{-4} erg.

Thus, fission of 1 nucleus of $^{235}_{92}$U releases about 3.2×10^{-4} erg of energy.

235 g of $^{235}_{92}$U contains = 6.023×10^{23} nuclei

1 g of $^{235}_{92}$U contains = $\frac{1}{235} \times 6.023 \times 10^{23} = 2.563 \times 10^{21}$ nuclei

Fission of 1 nuclei of $^{235}_{92}$U releases energy = 3.2×10^{-4} erg = 3.20×10^{-11} W s

Fission of 2.563×10^{21} nuclei of $^{235}_{92}$U releases energy $= 2.563 \times 10^{21} \times 3.2 \times 10^{-11}$ W s

$= 8.2 \times 10^{10}$ W s

$= 8.2 \times 10^{7}$ kWs

$= \frac{8.2 \times 10^{7}}{3600} = 2.3 \times 10^{4}$ kWh

$= \frac{2.3 \times 10^{4}}{24} = 0.96 \times 10^{3}$ kWd

Therefore, 1 g of U or Pu on fission would produce about 1 MW of power for one day. The same energy production requires more than 3 tons of coal or about 2300 litres of fuel oil per day.

4.8 NUCLEAR FUSION

Figure 1.4 of Chapter 1 shows that the binding energy per nucleon in the lightest nuclei is (like that in heaviest nuclei) less than that for nuclei of intermediate mass number. It implies that the sum of the masses of individual light nuclei is more than would be the mass of the nuclei formed by their fusion. Therefore, the combination of two of the lightest nuclei by a process of fusion is thus energetically advantageous.

However, fusing two lighter nuclei into one heavier nucleus is not so easy. Before light nuclei can combine, their mutual Coulomb repulsion must be overcome. In contrast fission induced by thermal neutrons has no such Coulomb barrier and thus very low energy neutrons can induce fission in heavy nuclei.

Suppose we wish to fuse two $^{20}_{10}$Ne nuclei to form $^{40}_{20}$Ca nucleus. The Coulomb barrier for two $^{20}_{10}$Ne nuclei is about 21 MeV. Therefore, we have to accelerate one $^{20}_{10}$Ne nucleus to at least 21 MeV and bombard it on a stationary $^{20}_{10}$Ne nucleus. Only then the two fuse together. The *Q*-value for this reaction is about 20.7 MeV. Thus, the total energy released is 21 MeV kinetic

energy plus 20.7 MeV Q-value or about 42 MeV. This means 21 MeV is supplied and 42 MeV comes out. This is not a promising situation. Accelerators accelerating $^{20}_{10}$Ne provide a beam current of about 10^{-6} A at about 20 MeV. The total power produced is 20×10^6 V $\times 10^{-6}$ A = 2 W. This small power is of no practical use.

Another approach is to heat a container of neon gas to very high temperature, so that thermal energy becomes larger than the Coulomb repulsion between the two interacting nuclei. This process requires a temperature of the order of 10^{11} K. If instead of $^{20}_{10}$Ne, we take hydrogen, the temperature required are of the order of 10^6–10^7 K.

Despite of the drawbacks, energy released in fusion is at present a subject of intensive research to perfect techniques of heating fusible nuclei so that large amount of power could be produced. It is fusion which powers sun and stars and is therefore ultimately responsible for the life on earth.

4.8.1 Energy Released in Fusion

Unlike fission, fusion is not a natural process on earth. Like we have spontaneous fission, spontaneous fusion does not exist. This is because of the limitation imposed due to the Coulomb barrier. However, once we overcome Coulomb barrier, fusion becomes a dominant mode as two fusing nuclei quickly reach a state of minimum energy.

The most elementary fusion reaction

$$p + p \rightarrow {}^{2}_{2}\mathrm{He}$$

is not possible, as $^{2}_{2}$He is not a bound system. Once it is formed, it will immediately break up into the constituents.

Other possible fusion reaction is:

$${}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + \gamma$$

The energy released (Q-value) of this reaction is 23.8 MeV. Other similar reactions are:

$${}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + n \qquad (Q = 3.3 \text{ MeV})$$

$${}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{1}\mathrm{H} + p \qquad (Q = 4.0 \text{ MeV})$$

These reactions are generally called *deuterium–deuterium reactions or D–D reactions*.

Another fusion reaction is

$$^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + n \qquad (Q = 17.6\ \mathrm{MeV})$$

This reaction is called *deuterium–tritium reaction* or *D–T reactions*, which provides mono-energetic neutrons of 14.1 MeV. The *Q*-value of this reaction is 17.6 MeV or 2.82 □ 10^{-5} ergs or 2.82 □ 10^{-12} W s.

Let us calculate the energy evolved in nuclear fusion, when 1 g of ^{2}H and ^{3}H undergoes fusion.

Approximately, 1 g of a mixture of ^{2}H and ^{3}H will contain about 0.5 g of ^{2}H and 0.5 g of ^{3}H.

Therefore, 0.5 g of ^{2}H or ^{3}H will contain nuclei = $\frac{0.5}{2}$ □ 6.023 □ 10^{23} = 1.5 □ 10^{23}

1 fusion produces power = 2.82 □ 10^{-12} W s

1.5 □ 10^{23} fusions produce power = 1.5 □ 10^{23} □ 2.82 □ 10^{-12} = 4.23 □ 10^{11} W s

$$= 4.23 \square 10^{8}\ \mathrm{kW\ s}$$

$$= \frac{4.23 \times 10^{8}}{3600 \times 24} = 4.89 \square 10^{3}\ \mathrm{kW\ d}$$

$$\sim 5\ \mathrm{MW\ d}$$

which is much higher compared to power produced in case of fission.

4.8.2 Hydrogen Burning and Solar Energy

Sun and most other stars consist of largely hydrogen. The first fusion reaction occurring in sun and other stars is

$$^{1}_{1}\mathrm{H} + ^{1}_{1}\mathrm{H} \rightarrow ^{2}_{1}\mathrm{H} + e^{+} + \nu \qquad (Q = 1.44\ \mathrm{MeV})$$

This reaction occurs via weak interaction discussed in Chapter 8, the cross-section for which is very small (□10^{-33} b) at proton mean energy of 1 keV which corresponds to central temperature of sun as 10^{7} K. This is the slowest

reaction occurring in the sun. It has been calculated that for 10^{18} protons, only 1 deuterium nucleus is formed per second. Since the number of deuterons is very small compared to protons (1 deuteron for 10^{18} protons); the next most likely reaction is

$$^{2}_{1}\mathrm{H} + ^{1}_{1}\mathrm{H} \rightarrow ^{3}_{2}\mathrm{He} + \gamma \qquad (Q = 5.49\ \mathrm{MeV})$$

The reaction of $^{3}_{2}$He with protons

$$^{3}_{2}\mathrm{He} + ^{1}_{1}\mathrm{H} \rightarrow ^{4}_{3}\mathrm{Li}$$

is not possible as $^{4}_{3}$Li is not a bound system, so it immediately breaks up into constituents. Further, $^{3}_{2}$He is unlikely to interact with $^{2}_{1}$H as density of $^{2}_{1}$H is small and as soon as $^{2}_{1}$H is formed, it interact with $^{1}_{1}$H to form $^{3}_{2}$He. $^{3}_{2}$He thus formed keeps on wandering till it finds another $^{3}_{2}$He and interacts with it as under.

$$^{3}_{2}\mathrm{He} + ^{3}_{2}\mathrm{He} \rightarrow ^{4}_{2}\mathrm{He} + 2^{1}_{1}\mathrm{H} + \gamma \qquad (Q = 12.86\ \mathrm{MeV})$$

In a nutshell, four protons are converted to helium

$$4^{1}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + 2e^{+} + 2\nu$$

with a total Q-value of 26.7 MeV. This cycle of fusion reactions is known as *proton–proton cycle.*

Heavier elements like $^{12}_{6}$C are also present in the interior of the sun. In the presence of heavy elements like ^{12}C following types of fusion reactions occur:

$$^{12}_{6}\mathrm{C} + ^{1}_{1}\mathrm{H} \rightarrow ^{13}_{7}\mathrm{N} + \gamma$$
$$^{13}_{7}\mathrm{N} \rightarrow ^{13}_{6}\mathrm{C} + e^{+} + \nu$$
$$^{13}_{6}\mathrm{C} + ^{1}_{1}\mathrm{H} \rightarrow ^{14}_{7}\mathrm{N} + \gamma$$
$$^{14}_{7}\mathrm{N} + ^{1}_{1}\mathrm{H} \rightarrow ^{15}_{8}\mathrm{O} + \gamma$$
$$^{15}_{8}\mathrm{O} \rightarrow ^{15}_{7}\mathrm{N} + e^{+} + \nu$$
$$^{15}_{7}\mathrm{N} + ^{1}_{1}\mathrm{H} \rightarrow ^{12}_{6}\mathrm{C} + ^{4}_{2}\mathrm{He}$$

The net result of these reactions is:

$$4\,{}_1^1\mathrm{H} \rightarrow {}_2^4\mathrm{He} + 2e^+ + 2\nu \qquad (Q = 24.7\ \mathrm{MeV})$$

This cycle of fusion reaction is known as *carbon cycle* or *CNO cycle*. During this cycle ${}_6^{12}\mathrm{C}$ is neither created nor destroyed. ${}_6^{12}\mathrm{C}$ acts as a catalyst to aid fusion process. The two positrons in the final reaction on annihilation give another 2 MeV energy. Thus, total energy released in CNO cycle is 26.7 MeV.

One may ask the question, why four hydrogen atoms do not directly combine to give one He atom rather than going through CNO cycle. The reason is that even at high temperature available inside the sun, protons do not have sufficient kinetic energy to overcome the Coulomb repulsion.

It is not possible to get the CNO cycle on the earth as temperature needed is very high. Experiments show that there is enough hydrogen in the sun to maintain this CNO cycle, and the temperature of the sun will remain practically unchanged for 10^9 years.

4.8.3 Helium Burning in Stars

As the age of the star advances, hydrogen at the core of the star continues to get depleted as it is converted into helium. Eventually, a helium core develops and continues to grow at the centre of the star, while energy due to hydrogen gets restricted to the outer shell surrounding the helium core. As the helium core grows, the star continues to contract releasing gravitational energy and raising the core temperature. Due to these conditions, the condition of the onset of helium burning at the core is attained. For this to occur, mean temperature of the helium core should be of the order of 10^8 K. At this temperature again a state of equilibrium in the star is reached and this burning can continue for 10^8 years. This helium burning reaction, also known as triple alpha reaction takes place in two steps:

$${}_2^4\mathrm{He} + {}_2^4\mathrm{He} \rightarrow {}_4^8\mathrm{Be} + \gamma \qquad (Q = -0.095\ \mathrm{MeV})$$

$${}_4^8\mathrm{Be} + {}_2^4\mathrm{He} \rightarrow {}_6^{12}\mathrm{C} + \gamma \qquad (Q = 7.4\ \mathrm{MeV})$$

First reaction is endothermic and second reaction is exothermic. Overall energy released in helium burning is 7.3 MeV, which is much less than that released in hydrogen burning.

NUMERICAL PROBLEMS

Section 4.2

Solved Problems

1. Complete the following reactions:

(i) $^{59}_{27}\text{Co} + ^{1}_{0}n \rightarrow ? + \gamma$ (ii) $^{14}_{7}\text{N} + ^{1}_{1}\text{H} \rightarrow ^{11}_{6}\text{C} + ?$

(iii) $^{64}_{30}\text{Zn} + ^{1}_{0}n \rightarrow ? + ^{1}_{0}n + ^{1}_{0}n$ (iv) $^{238}_{92}\text{U} + ^{14}_{7}\text{N} \rightarrow ^{243}_{97}\text{Bk} + 5^{1}_{0}n + ?$

Solution:

(i) Here total number of protons of the reactants is 27 and total number of nucleons is 60. Therefore, the resultant nucleus will have 27 protons and 60 nucleons. So, the missing nucleus is $^{60}_{27}\text{Co}$ and the reaction is

$$^{59}_{27}\text{Co} + ^{1}_{0}n \rightarrow ^{60}_{27}\text{Co} + \gamma$$

Similarly, for other reactions.

(ii) The missing particle is $^{4}_{2}\text{He}$ and the complete reaction is

$$^{14}_{7}\text{N} + ^{1}_{1}\text{H} \rightarrow ^{11}_{6}\text{C} + ^{4}_{2}\text{He}$$

(iii) The missing nucleus is $^{62}_{30}\text{Zn}$ and the complete reaction is

$$^{64}_{30}\text{Zn} + ^{1}_{0}n \rightarrow ^{63}_{30}\text{Zn} + ^{1}_{0}n + ^{1}_{0}n$$

(iv) The missing particle is $^{4}_{2}\text{He}$ and the complete reaction is

$$^{238}_{92}\text{U} + ^{14}_{7}\text{N} \rightarrow ^{243}_{97}\text{Bk} + 5^{1}_{0}n + ^{4}_{2}\text{He}$$

2. The ratio of activities of $\frac{^{14}\text{C}}{^{32}\text{P}}$ today at $t = 0$ is 1. What will be the ratio after 14 days and after 70 days? Given half-life of ^{14}C = 5700 years and that of ^{32}P = 14 days

Solution: Let $A_{0\text{C}}$, and $A_{0\text{P}}$ be the initial activities of ^{14}C and ^{32}P, respectively.

Therefore,

$$\frac{A_{0C}}{A_{0P}} = 1$$

We have ^{14}C

$$A_C = A_{0C} e^{-\lambda C t}$$

and for ^{32}P

$$A_P = A_{0P} e^{-\lambda P t}$$

Taking the ratio, we get

$$\frac{A_C}{A_P} = \frac{A_{0C} e^{-\lambda C t}}{A_{0P} e^{-\lambda P t}}$$

Given $\frac{A_{0C}}{A_{0P}} = 1$

and

$$\lambda_C = \frac{0.6931}{5700 \times 365 \times 24 \times 3600} = 3.86 \times 10^{-12} s^{-1}$$

$$\lambda_P = \frac{0.6931}{14 \times 24 \times 3600} = 0.57 \times 10^{-6} s^{-1}$$

Therefore, after 14 days

$$\frac{A_C}{A_P} = \frac{e^{-3.86 \times 10^{-12} \times 14 \times 24 \times 3600}}{e^{-0.57 \times 14 \times 24 \times 3600}}$$

$$\frac{A_C}{A_P} \approx 2$$

and after 70 days

$$\frac{A_C}{A_P} = \frac{e^{-3.86 \times 10^{-12} \times 70 \times 24 \times 3600}}{e^{-0.57 \times 70 \times 24 \times 3600}}$$

$$\frac{A_C}{A_P} \approx 31$$

Unsolved Problems

1. Find the missing nucleus/particle in the following reactions:

(i) ${}^{31}_{15}\mathrm{P} + \gamma \rightarrow ? + {}^{1}_{0}n$

(ii) ${}^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{30}_{15}\mathrm{P} + ?$

(iii) ${}^{59}_{27}\mathrm{Co} + {}^{4}_{2}\mathrm{He} \rightarrow ? + 2{}^{1}_{0}n$

(iv) ${}^{59}_{27}\mathrm{Co} + {}^{3}_{2}\mathrm{He} \rightarrow ? + {}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H}$

(v) ${}^{27}_{13}\mathrm{Al} + {}^{3}_{1}\mathrm{H} \rightarrow ? + {}^{2}_{1}\mathrm{H}$

(vi) ${}^{27}_{13}\mathrm{Al} + ? \rightarrow {}^{26}_{13}\mathrm{Al} + {}^{3}_{1}\mathrm{H}$

(vii) $? + {}^{1}_{0}n \rightarrow {}^{63}_{30}\mathrm{Zn} + {}^{1}_{0}n + {}^{1}_{0}n$

(viii) ${}^{24}_{12}\mathrm{Mg} + {}^{1}_{0}n \rightarrow {}^{25}_{12}\mathrm{Mg} + ?$

(ix) ${}^{197}_{79}\mathrm{Au} + {}^{12}_{6}\mathrm{C} \rightarrow {}^{205}_{85}\mathrm{At} + ?$

(x) ${}^{27}_{13}\mathrm{Al} + {}^{14}_{7}\mathrm{N} \rightarrow {}^{25}_{13}\mathrm{Al} + ?$

[**Ans.** (i) ${}^{30}_{15}\mathrm{P}$, (ii) ${}^{1}_{0}n$, (iii) ${}^{61}_{29}\mathrm{Cu}$, (iv) ${}^{60}_{27}\mathrm{Co}$, (v) ${}^{28}_{13}\mathrm{Al}$, (vi) ${}^{2}_{1}\mathrm{H}$, (vii) ${}^{64}_{30}\mathrm{Zn}$, (viii) γ, (ix) $4{}^{1}_{0}n$, (x) ${}^{16}_{7}\mathrm{N}$]

2. ${}^{27}\mathrm{Al}$ is bombarded with neutrons. Write the following reactions:

$$(n, p), (n, a), (n, d), (n, g)$$

3. Complete the following nuclear reactions:

(i) ${}^{27}\mathrm{Al}(a, p)$

(ii) ${}^{27}\mathrm{Al}(a, d)$

(iii) ${}^{27}\mathrm{Al}(a, a)$

(iv) ${}^{27}\mathrm{Al}(a, n)$

(v) ${}^{27}\mathrm{Al}(a, g)$

(vi) ${}^{27}\mathrm{Al}(a, {}^{3}\mathrm{He})$

(vii) ${}^{27}\mathrm{Al}(a, {}^{3}\mathrm{H})$

4. A neutron and a proton can undergo radioactive capture at rest:

$$p + n \rightarrow d + \gamma$$

Find the energy of the photon emitted in this capture
m_p = 1.00783 amu, m_n = 1.00867 amu, m_d = 2.01410 amu [**Ans.** 2.235 MeV]

5. Deuterons undergo photodisintegration reaction with g-rays of energy 2.225 MeV. Given m_d = 2.0141022 amu, m_p = 1.0078252 amu and 1 amu = 931.5 MeV, find the mass of neutron. [**Ans.** 1.0086656 amu]

Section 4.3

Solved Problems

1. A 0.01 mm thick $^{7}_{3}$Li target is bombarded with a beam of flux of 10^{13} particles/cm^2-s. As a result 10^8 neutrons/s are produced. Calculate the cross-section for this reaction. Given: density of lithium = 500 kg/m^3.

Solution: Thickness of $^{7}_{3}$Li = t = 0.01 mm = 10^{-5} m

Density = 500 kg/m^3

$$\text{Number of } {}^{7}_{3}\text{Li nuclei/volume } n = \frac{\rho N}{M}$$

$$= \frac{500 \times 6.023 \times 10^{26}}{7}$$

$$= 4.302 \times 10^{28}/\text{m}^3$$

Number of $^{7}_{3}$Li nuclei/area = $4.302 \times 10^{28} \times t$

$$= 4.302 \times 10^{28} \times 10^{-5}$$

$$= 4.302 \times 10^{23}$$

Number of nuclei undergoing interactions/s = Number of neutrons produced/s

Number of neutrons produced/s = 10^8

Number of incident particles striking/unit area of target N_0 = 10^{13}/unit area of target.

$$\text{Cross-section } \sigma = \frac{\text{Number of neutrons produced/s}}{N_0 \times n}$$

$$= \frac{10^8}{10^{13} \times 4.302 \times 10^{23}}$$

$$= 2.32 \times 10^{-29}\ \text{m}^2$$

$$= 0.232\ \text{b}\ (1\ \text{b} = 10^{-28}\ \text{m}^2)$$

2. Neutron capture cross-section of thermal neutrons for ^{113}Cd is 20,000 b. The mean atomic mass of natural cadmium is 112 amu and its density is 8.64×10^3 kg/m^3. What fraction of an incident beam of thermal neutrons is absorbed by a cadmium sheet of thickness 0.2 mm? Given abundance of ^{113}Cd in natural cadmium is 12%.

Solution: Number of natural cadmium atoms/volume = $\frac{r\,N}{M}$

Thus, number of ^{113}Cd atoms present in natural cadmium

$$n = 0.12\,\frac{6.023\times10^{26}\times8.64\times10^{3}}{112}$$

or

$$n = 5.58 \square 10^{27} \text{ atoms/m}^3.$$

Now,

$$s = 20{,}000 = 20{,}000 \square 10^{-28} = 2 \square 10^{-24} \text{ m}^2$$

and

$$ns = 5.58 \square 10^{27} \square 2 \square 10^{-24} = 1.16 \square 10^{4} \text{ m}^{-1}$$

Now, we have $N = N_0 e^{-n}s^{x}$

Therefore, the fraction of incident neutrons absorbed is

$$\frac{N_0 - N}{N_0} = \frac{N_0 - N_0 e^{-n\sigma x}}{N_0} = 1 - e^{-n\sigma x}$$

$$x = 0.2 \text{ mm} = 2\times10^{-4} \text{ m}$$

$$\frac{N_0 - N}{N_0} = 1 - e^{-1.16\times10^{4}\times2\times10^{-4}} = 0.894$$

So, 89.4% of incident neutrons is absorbed in the sheet of natural cadmium.

3. A beam of thermal neutrons while traversing a 2 mm thick foil of ^{197}Au, some 70% of the neutrons are removed. What is the total thermal neutron cross-section? Comment on result obtained on cross–section, given that the radius of a gold nucleus is 6.5 $\square$ 10^{-15} m. (Density of gold: 19,300 kg m^{-3}.)

Solution: To solve this problem, we use the equation

$$N(x) = N_0 e^{-s\,nx}$$

where s is the cross-section, n is the number of absorbing or scattering nuclei per unit volume and x is the thickness of the absorber. The number of nuclei per unit volume is given by

$$n = \frac{N_{av} r}{A}$$

where N_{av} is Avogadros number, r is the density and A is the atomic mass number for gold. Putting the numbers give s = 102 barns. Assuming this cross-section as area, or area = pr^2, which gives $r = \sqrt{\frac{\text{area}}{p}}$.

Interpreting this as an area yields a radius of 5.7 □ 10^{-14} m which is almost nine times larger than the actual radius given. This indicates that resonance capture is occurring.

Unsolved Problems

1. A thin sheet of ^{59}Co, 0.05 cm thick is irradiated with a neutron beam of flux 10^{12} n/cm^2/s for 10 hours. If the cross-section of the neutron capture by ^{59}Co is 37 b, how many nuclei of the isotope ^{60}Co would be produced at the end of irradiation/cm^3 and what is the initial b^--activity of the sample? $t_{1/2}$ for ^{60}Co = 5.2 years and density of cobalt = 8.9 g/cc. [**Ans.** 6.5 □ 10^{15}, 690 □Ci]
2. The capture cross-section of ^{59}Co for thermal neutrons (energy = 0.025 eV) is 37 b. What percentage of a beam of thermal neutrons will penetrate a 2.0 mm sheet of ^{59}Co? Density of cobalt = 8.9 □ 10^3 kg/m^3. [**Ans.** 51%]
3. The mean free path of neutrons in bismuth is about 6.5 cm. Find the total neutron cross- section of bismuth. (Atomic number ~ 209, density ~9.8 g/cm^3.) [**Ans.** 5.46 b]
4. Find the mean free path of neutrons in bismuth. Given the total neutron cross-section of bismuth 5.46 barns. (Atomic number ~209, density ~9.8 g/cm^3) [**Ans.** 6.5 cm]

Section 4.4

Solved Problem

1. Are the following reactions/decays possible? If not, write the correct relation.

(i) ${}^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{30}_{14}\mathrm{Si} + {}^{1}_{0}n$ (ii) ${}^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{143}_{56}\mathrm{Ba} + {}^{90}_{36}\mathrm{Kr} + 2{}^{1}_{0}n$

(iii) ${}^{32}_{15}\mathrm{P} \rightarrow {}^{32}_{16}\mathrm{S} + {}^{0}_{-1}e + \nu_e$

Solution:

(i) ${}^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{30}_{14}\mathrm{Si} + {}^{1}_{0}n$

In the reaction, the number of protons in the reactants is 13 + 2 = 15. In the products, the number of protons is 14 + 0 = 14. Hence, number of nucleons is not conserved. In the products, ${}^{1}_{0}n$ should be replaced by ${}^{1}_{1}\mathrm{H}$. The correct equation is

${}^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{30}_{14}\mathrm{Si} + {}^{1}_{1}\mathrm{H}$

(ii) ${}^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{143}_{56}\mathrm{Ba} + {}^{90}_{36}\mathrm{Kr} + 2{}^{1}_{0}n$

In the reaction, the number of nucleons in the reactants is 235 + 1 = 236. In the products, the number of nucleons is 143 + 90 + 2 = 235. Hence, number of nucleons is not conserved. In the products, $2\,{}^{1}_{0}n$ should be replaced by $3\,{}^{1}_{0}n$. The correct equation is

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{143}_{56}\mathrm{Ba} + {}^{90}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}n$$

(iii) $$^{32}_{15}\mathrm{P} \rightarrow {}^{32}_{16}\mathrm{S} + {}^{0}_{-1}e + \nu_e$$

In the reaction, the number of leptons in the reactants is 0 + 0 = 0. In the products, the number of leptons is 1 + 1 = 2. Hence, lepton number is not conserved. In the products, n_e should be replaced by $\bar{n}_e$. The correct equation is

$$^{32}_{15}\mathrm{P} \rightarrow {}^{32}_{16}\mathrm{S} + {}^{0}_{-1}e + \bar{\nu}_e$$

Unsolved Problem

1. Are the following reactions/decays possible? If not, write the correct relation.

(i) $^{137}_{55}\mathrm{Cs} \rightarrow {}^{136}_{56}\mathrm{Ba} + {}^{0}_{-1}e + \bar{\nu}_e$

(ii) $^{60}_{27}\mathrm{Co} \rightarrow {}^{60}_{28}\mathrm{Ni} + {}^{0}_{-1}e$

(iii) $^{136}_{55}\mathrm{Ba}^{*} \rightarrow {}^{136}_{56}\mathrm{Ba} + \gamma$

(iv) $^{22}_{11}\mathrm{Na} \rightarrow {}^{22}_{10}\mathrm{Ne} + {}^{0}_{-1}e$

(v) $^{24}_{12}\mathrm{Mg} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{25}_{10}\mathrm{Ne} + {}^{1}_{1}\mathrm{H}$

(vi) $^{4}_{2}\mathrm{He} + {}^{9}_{4}\mathrm{Be} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{1}_{0}n$

(vii) $^{115}_{49}\mathrm{In} + {}^{6}_{3}\mathrm{Li} \rightarrow {}^{118}_{50}\mathrm{Sn} + {}^{3}_{1}\mathrm{H}$

(viii) $^{197}_{79}\mathrm{Au} + {}^{28}_{14}\mathrm{Si} \rightarrow {}^{203}_{85}\mathrm{At} + {}^{15}_{8}\mathrm{O}$

Section 4.5

Solved Problems

1. Calculate the Q-value for the reaction

$$^{2}\mathrm{H} + {}^{3}\mathrm{H} \square {}^{4}\mathrm{He} + {}^{1}n$$

Given:

$M(^{1}n) = 1.00866501$ amu

$M(^{2}\mathrm{H}) = 2.014102$ amu

$M(^{3}\mathrm{H}) = 3.016049$ amu

$M(^4\text{He}) = 4.002603$ amu

Solution: Q-value is given by

$$Q = [M(^2\text{H}) + M(^3\text{H}) - M(^4\text{He}) - M(^1n)]$$
$$= [2.014102 + 3.016049 - 4.002603 - 1.00866501]$$
$$= 0.01888299 \text{ amu}$$
$$= 0.01888299 \square 931.49 = 17.6 \text{ MeV}$$

Therefore, the Q-value for this reaction is 17.6 MeV.

2. Prove that ^{252}Cf is unstable and decays by a-emission as under:

$$^{252}\text{Cf} \square {}^{248}\text{Cm} + {}^4\text{He}$$

Given:

$M(^{252}\text{Cf}) = 252.081621$ amu

$M(^{248}\text{Cm}) = 248.072343$ amu

$M(^4\text{He}) = 4.002603$ amu

Solution: Let us calculate the Q-value for this decay

$$Q = [M(^{252}\text{Cf}) - M(^{248}\text{Cm}) - M(^4\text{He})]$$
$$= [252.081621 - 248.072343 - 4.002603]$$
$$= 0.006675 \text{ amu}$$
$$= 0.006675 \square 931.49 = 6.21 \text{ MeV}$$

Since Q-value for this reaction is +6.21 MeV, therefore, ^{252}Cf will decay by a-emission.

3. Find out the Q-value for the reaction

$$^{208}\text{Pb}(^{56}\text{Fe}, {}^{54}\text{Fe})^{210}\text{Pb}$$

Given:

$M(^{208}\text{Pb}) = 207.976641$ amu

$M(^{56}\text{Fe}) = 55.934939$ amu

$M(^{210}\text{Pb}) = 209.984178$ amu

$M(^{54}\text{Fe}) = 53.939612$ amu

Also find the threshold for this reaction.

Solution: In this reaction the projectile is ^{56}Fe and the reaction is

$$^{208}\text{Pb} + {}^{56}\text{Fe} \rightarrow {}^{210}\text{Pb} + {}^{54}\text{Fe}$$

the Q-value for this reaction is given by

$$\begin{aligned} Q &= [M(^{208}\text{Pb}) + M(^{56}\text{Fe}) - M(^{210}\text{Pb}) - M(^{54}\text{Fe})] \text{ amu} \\ &= [207.976641 + 55.934939 - 209.984178 - 53.939612] \text{ amu} \\ &= -0.001221 \text{ amu} = -0.01221 \times 931.49 \text{ MeV} = -11.37 \text{ MeV} \end{aligned}$$

Therefore, the Q-value for this reaction is –11.37 MeV.

Threshold for this reaction is

$$E_{\text{th}} = -Q \frac{M(\text{projectile}) + M(\text{target})}{M(\text{target})}$$

or

$$= -Q \frac{M(^{56}\text{Fe}) + M(^{208}\text{Pb})}{M(^{208}\text{Pb})}$$

In this relation atomic masses can be replaced by respective mass numbers. The error introduced by this is negligible.

Therefore,

$$E_{\text{th}} = 11.37 \frac{56 + 208}{208}$$

$$= 14.43 \text{ MeV}$$

So, the threshold for the reaction is 14.43 MeV.

4. When a proton captures a neutron to produce a deuteron nucleus, a g-ray of energy 2.230 MeV is released based on the following equation:

$$^{1}_{1}\text{H} + {}^{1}_{0}n \rightarrow {}^{2}_{1}\text{H} + \gamma$$

Calculate the mass of neutron. Given:

Mass of $^{1}_{1}$H = 1.008142 amu

Mass of $^{2}_{1}$H = 2.014735 amu

Solution: Total mass of the reactants = Total mass of the products

Let us take all masses in MeV

Therefore,

$$1.008142 \square 931.47 + m \square 931.47 = 2.014735 \square 931.47 + 2.230$$

where m is the mass of neutron.

$$931.47 \times m = 939.8613$$

$$m = \frac{939.8412}{931.47} = 1.008987 \text{ amu}$$

5. Find out whether the following reaction is exoergic or endoergic.

$$^{6}\text{Li} + {}^{1}n \square {}^{4}\text{He} + {}^{3}\text{H}$$

Given:

$M(^{6}\text{Li}) = 6.0151234$ amu

$M(^{1}n) = 1.0086654$ amu

$M(^{4}\text{He}) = 4.0026034$ amu

$M(^{3}\text{H}) = 3.0160294$ amu

Solution: Q-value for this reaction is given by

$$Q = [M(^{6}\text{Li}) + M(^{1}n) - M(^{4}\text{He}) - M(^{3}\text{H})]$$
$$= [6.0151234 + 1.0086654 - 4.0026034 - 3.0160294]$$
$$= 0.005156 \text{ amu}$$
$$= 0.005156 \square 931.47 = 4.8 \text{ MeV}$$

Therefore, the Q-value for this reaction is 4.8 MeV and the reaction is exoergic.

6. Prove that ^{252}Cf can undergo spontaneous fission as under:

$$^{252}\text{Cf} \square {}^{98}\text{Zr} + {}^{145}\text{Ce} + 9{}^{1}n$$

Given:

$M(^{252}\text{Cf}) = 252.081621$ amu

$M(^{98}\text{Zr}) = 97.912735$ amu

$M(^{145}\text{Ce}) = 144.917230$ amu

$M(^{1}n) = 1.008664916$ amu

Solution: Q-value for the spontaneous fission is given by

$$Q = M(^{252}\text{Cf}) - M(^{98}\text{Zr}) - M(^{145}\text{Ce}) - 9 \square M(^{1}n)$$
$$= (252.081621 - 97.912735 - 144.917230 - 9.0779850)$$
$$= 0.1737009 \text{ amu}$$
$$= 0.1737009 \square 931.47 = 161.8 \text{ MeV}$$

Since Q-value is positive, so spontaneous decay of ^{252}Cf is possible.

7. Calculate the separation energy to remove one proton from $^{16}_{8}$O. Given:

$$m(\text{H}) = 1.007825 \text{ amu}$$
$$m(^{15}_{7}\text{N}) = 15.000108 \text{ amu}$$
$$m(^{16}_{8}\text{O}) = 16 \text{ amu}$$

Solution: Given reaction is

$$^{16}_{8}\text{O} \rightarrow {}^{15}_{7}\text{N} + {}^{1}_{1}\text{H}$$

Q-value for this reaction is

$$Q = M_{\text{O}} - M_{\text{N}} - M_{\text{H}}$$
$$= 16 - 15.000108 - 1.007825 = -0.007933$$
$$= -0.007933 \square 931.47$$
$$= -7.4 \text{ MeV}$$

Therefore, energy required to remove one proton from $^{16}_{8}$O = 7.4 MeV

8. The Q-value of the ^{23}Na$(n, a)^{20}$F reaction is –5.4 MeV. Determine the threshold energy of the neutrons for this reaction. Given:

Mass of neutron = 1.00866 amu

Mass of ^{23}Na = 22.99097 amu

Solution: Given reaction is

$$^{23}\text{Na} + n \rightarrow {}^{20}\text{F} + {}^{4}_{2}\text{He}$$
$$Q\text{-value} = -5.4 \text{ MeV}$$
$$\text{Therefore, threshold energy} = -Q\frac{M_{\text{Na}} + M_n}{M_{\text{Na}}}$$
$$= -5.4\,\frac{22.99097 + 1.00866}{22.99097}$$
$$= 5.64 \text{ MeV.}$$

9. Calculate the *Q*-value in MeV for the following nuclear reaction:

$$^{4}_{2}\mathrm{He} + {}^{14}_{7}\mathrm{N} \rightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{H}$$

Given masses of

$$^{4}_{2}\mathrm{He} = 4.00388 \text{ amu}$$
$$^{14}_{7}\mathrm{N} = 14.00755 \text{ amu}$$
$$^{17}_{8}\mathrm{O} = 17.00452 \text{ amu}$$
$$^{1}_{1}\mathrm{H} = 1.00815 \text{ amu}$$

Solution: *Q*-value is given by the relation

$$Q = M_{\mathrm{N}} + M_{\mathrm{He}} - M_{\mathrm{O}} - M_{\mathrm{H}}$$
$$= 14.00755 + 4.00388 - 17.00452 - 1.00866$$
$$= -0.00124 \text{ amu}$$
$$= -0.00124 \times 931.47 = -1.15 \text{ MeV}$$

Therefore, *Q*-value = –1.15 MeV

10. Calculate the minimum energy of *g*-rays necessary to disintegrate deuteron into a proton and neutron. Given masses of

$$^{2}_{1}\mathrm{H} = 2.014735 \text{ amu}$$
$$^{1}_{1}\mathrm{H} = 1.008142 \text{ amu}$$
$$^{1}_{0}n = 1.008987 \text{ amu}$$

Solution: The given reaction is

$$^{2}_{1}\mathrm{H} + g \rightarrow {}^{1}_{1}\mathrm{H} + {}^{1}n$$

Let *Eg* be the energy of *g*-ray to start this reaction.
Total mass of the reactants = Total mass of the products
Let us take all masses in MeV
Therefore,

$$2.014735 \times 931.47 + Eg = 1.008142 \times 931.47 + 1.008987 \times 931.47$$

or

$$Eg = 2.2299 \text{ MeV}$$

Unsolved Problems

1. Find out whether the following reaction is exoergic or endoergic.

$$^{27}\text{Al} + {}^{1}\text{H} \square {}^{27}\text{Si} + {}^{1}n$$

Given:

$M(^{1}\text{H}) = 1.00727647$ amu

$M(^{1}n) = 1.008664916$ amu

$M(^{27}\text{Al}) = 26.981539$ amu

$M(^{27}\text{Si}) = 26.986704$ amu [**Ans.** $Q = -6.10$ MeV, endoergic]

2. Find out the Q-value for the reaction

$$^{56}\text{Fe}(^{208}\text{Pb}, {}^{210}\text{Pb})^{54}\text{Fe}$$

Given:

$M(^{208}\text{Pb}) = 207.976641$ amu

$M(^{56}\text{Fe}) = 55.934939$ amu

$M(^{210}\text{Pb}) = 209.984178$ amu

$M(^{54}\text{Fe}) = 53.939612$ amu

Also find the threshold for this reaction. [**Ans.** –11.37 MeV, 54.60 MeV]

3. (i) Calculate the Q-value for the decay of a neutron into a proton and an electron. [**Ans.** 0.783 MeV□

(ii) Calculate the Q-value for the decay of a proton into a neutron and an electron. [**Ans.** –1.805 MeV]

4. Calculate the Q-value for the reaction

$$^{27}\text{Al} + {}^{4}\text{He} \square {}^{30}\text{Si} + {}^{1}\text{H}$$

Given:

$M(^{1}\text{H}) = 1.00727647$ amu

$M(^{4}\text{He}) = 4.002603$ amu

$M(^{27}\text{Al}) = 26.981539$ amu

$M(^{30}\text{Si}) = 29.973772$ amu [**Ans.** 2.88 MeV]

5. Find the minimum energy of *g*-ray required to produce photodisintegration of ^{9}Be.

Given:

$M(^{9}\text{Be}) = 9.012182$ amu

$M(^{4}\text{He}) = 4.002603$ amu

$M(^{1}n) = 1.00866501$ amu [**Ans.** 1.57 MeV]

6. Find the minimum kinetic energy in the lab system needed by an *a*-particle to cause the reaction $^{14}\text{N}(a, p)^{17}\text{O}$. The masses of

$m(^{14}\text{N}) = 14.00307$ amu

$m(a) = 4.003879$ amu

$m(p) = 1.00783$ amu

$m(^{17}\text{O}) = 17.00013$ amu [**Ans.** 1.21 MeV]

7. Calculate the *Q*-value for the reaction

$$^{27}\text{Al} + ^{1}\text{H} \rightarrow ^{27}\text{Si} + ^{1}n$$

Given:

$M(^{1}\text{H}) = 1.00727647$ amu

$M(^{1}n) = 1.008664916$ amu

$M(^{27}\text{Al}) = 26.981539$ amu

$M(^{27}\text{Si}) = 26.986704$ amu [**Ans.** –6.10 MeV]

8. Find the *Q*-value and threshold for the reaction

$$^{239}\text{U}(^{4}\text{He}, 3^{1}n)^{239}\text{Pu}$$

Given:

$M(^{238}\text{U}) = 238.050785$ amu

$M(^{4}\text{He}) = 4.002603$ amu

$M(^{239}\text{Pu}) = 239.052158$ amu

$M(^{1}n) = 1.00866501$ amu [**Ans.** 23.1 MeV, 23.5 MeV]

9. Calculate the threshold energy for $^{14}\text{N}(n, a)^{11}\text{B}$ reaction. Given:

$m(n) = 1.008983$ amu

$m(a) = 4.003879$ amu

$m(^{14}\text{N}) = 14.007550$ amu

$m(^{11}\text{B}) = 11.012811$ amu [**Ans.** 0.157 MeV]

10. Find the Q-value for the nuclear reaction

$$^{9}_{4}\text{Be} + ^{4}_{2}\text{He} \rightarrow ^{12}_{6}\text{C} + ^{1}_{0}n + Q$$

Given masses of

$^{9}_{4}\text{Be} = 9.01286$ amu

$^{4}_{2}\text{He} = 4.00388$ amu

$^{12}_{6}\text{C} = 12.000$ amu

$^{1}_{0}n = 1.00866$ amu [**Ans.** 7.52 MeV]

11. A tritium gas target $(^{3}_{1}\text{H})$ is bombarded with a beam of protons $(^{1}_{1}\text{H})$ of kinetic energy 3 MeV. Determine the Q-value of the following reaction:

$^{1}_{1}\text{H} + ^{3}_{1}\text{H} \square ^{3}_{2}\text{He} + ^{1}_{0}n + Q$

Given:

$m(^{1}_{1}\text{H}) = 1.007276$ amu

$m(^{3}_{1}\text{H}) = 3.016056$ amu

$m(^{1}_{0}n) = 1.00866$ amu

$m(^{3}_{2}\text{He}) = 3.016030$ amu [**Ans.** –1.26 MeV]

12. A beam of a-particles of energy 7.3 MeV is used for $^{27}\text{Al}\ (a, p)^{30}\text{Si}$ reaction. The protons emitted at an angle of 0° are found to have energy of 9.34 MeV. Find the Q-value of this reaction. [**Ans.** 2.22 MeV]

13. A deuteron beam of energy 1.51 MeV is used for $^{16}\text{O}(d, a)\ ^{14}\text{N}$ reaction. a-particles emitted at an angle of 90° are found to have energy of 3.42 MeV. Find the Q-value of this reaction. [**Ans.** 3.10 MeV]

Section 4.6

Unsolved Problems

1. What target isotope must be used to form the compound nucleus ^{28}Si, when the projectile is (i) neutron, (ii) proton, and (iii) alpha-particle? [**Ans.** (i) ^{27}Si, (ii) ^{27}Al, (iii) ^{24}Mg]
2. List at least four nuclear reactions for making compound nucleus ^{36}Cl by bombarding different projectiles on stable nuclei.
3. When ^{27}Al is bombarded with protons, what compound nucleus is formed? List four possible decay modes of this compound nucleus assuming that sufficient energy exists with the compound nucleus for decay into the chosen mode.
4. What target isotope must be used to form the compound nucleus ^{24}Na, when the projectile is (i) n, (ii) p, (iii) a-particle and (iv) ^{3}He?
5. Devise three nuclear reactions that forms ^{24}Mg as a compound nucleus starting with stable target and stable projectile.
6. List at least three nuclear reactions for making compound nucleus ^{60}Co by bombarding different projectiles on stable nuclei.

Section 4.7

Solved Problems

1. ^{235}U loses about 0.1% of its mass when it undergoes fission.
   (i) How much energy is released when 1 kg of ^{235}U undergoes fission?
   (ii) One ton of TNT releases about 4 GJ when it is detonated. How many tons of TNT are equivalent in destructive power to a bomb that contains 1 kg of ^{235}U?

Solution: (i) 0.1% of 1 kg = 0.001 kg

Applying the relation $E = mc^2$, we get

$E = 0.001 \times (3 \times 10^8)^2 = 9 \times 10^{13}$ J

(ii) 4 GJ = 4×10^9 J of energy is produced by 1 t of TNT

9×10^{13} J of energy requires TNT = $\frac{1}{4 \times 10^9} \times 9 \times 10^{13} = 22{,}500$ t

Therefore, the destructive power of bomb using 1 kg of ^{235}U = 22,500 t of TNT = 22.5 kt of TNT.

2. A beam of low energy neutrons, intensity $10^6 s^{-1}$ traverses a foil of ^{235}U, thickness

0.15 kg m^{-2}. If the fission cross-section is 2 □ $10^{-26} m^2$, find the rate of fissions induced in the foil by the neutrons. Here, we neglect the reduction of the beam as it passes through the foil.

Solution: Thickness of the foil = 0.15 kg m^{-2}

This means that a foil of area S m^2 will have a mass of S □ 0.15 kg.

235 kg of ^{235}U contains = 6.023 □ 10^{26} nuclei

S □ 0.15 kg will contain = $\frac{S \times 0.15 \times 6.023 \times 10^{26}}{235}$

$$= S \square 3.844 \square 10^{23} \text{ nuclei}$$

Each nucleus presents an area = 2 □ $10^{-26} m^2$ to the beam of neutrons

So the total area presented = S □ 3.844 □ 10^{23} □ 2 □ $10^{-26} m^2$.

$$= S \square 7.686 \square 10^{-3}$$

Thus, the fraction of the area S which is covered = 7.686 □ 10^{-3}

This is the fraction of the beam removed by the fission process. Hence, the total induced fission rate in this fractional area is 7.686 □ 10^{-3} □ 10^6 = 7685 s^{-1}, when a beam of intensity $10^6 s^{-1}$ is incident on that area.

3. The nuclide ^{256}Fm decays through spontaneous fission with a half-life of 158 min. If the energy released is about 220 MeV per fission, calculate the fission power produced by 1 □g of this isotope.

Solution: We have the fission rate

$$\frac{dN}{dt} = -Nl$$

Here, N is the number of nuclei present and l is the decay constant.

Number of nuclei of ^{256}Fm present in 1 □g of sample is

$$= \frac{6.023 \times 10^{23}}{256} \times 10^{-6} = 2.35 \times 10^{15}$$

and decay constant $l = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{158 \times 60} = 7.31 \times 10^{-5} \text{s}^{?1}$

Therefore, fission rate = 2.35 □ 10^{15} □ 7.31 □ 10^{-5}

= 1.72 □ 10^{11} fissions s^{-1}

Energy released in fission of one nucleus = 220 MeV

= 220 □ 1.602 □ 10^{-13} J

= 3.52 □ 10^{-11} J

Power released in fission of 1.72 □ 10^{11} nuclei = 3.52 □ 10^{-11} □ 1.72 □ 10^{11}

= 6 W

4. The nuclide ^{252}Cf undergoes spontaneous fission with a half-life of 2.62 years. Energy released in this process is about 210 MeV. If 100 mg of this isotope is taken in a space- ship, calculate the power produced by this isotope.

Solution: We have the fission rate

$$\frac{dN}{dt} = -Nl$$

Here, N is the number of nuclei present and l is the decay constant.

Number of nuclei of ^{252}Cf present in 100 mg of sample is

$$= \frac{6.023 \times 10^{23}}{252} \times 0.1 = 2.39 \times 10^{20}$$

and decay constant $l = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{2.62 \times 365 \times 24 \times 3600} = 8.39 \times 10^{-9} \text{s}^{-1}$

Therefore, fission rate = 2.39 □ 10^{20} □ 8.39 □ 10^{-9}

= 2.01 □ 10^{12} fissions/s

Energy released in fission of one nucleus = 210 MeV

$$= 210 \times 1.602 \times 10^{-13} \text{ J}$$

$$= 3.36 \times 10^{-11} \text{ J}$$

Energy released in fission of 2.01×10^{12} nuclei = $3.36 \times 10^{-11} \times 2.01 \times 10^{12}$

$$= 67.6 \text{ W}$$

5. The energy released during the nuclear explosion at Hiroshima has been estimated as equivalent to that released by 20,000 tons of TNT. Assume that 200 MeV is released when a ^{235}U nucleus absorbs a neutron and fissions and that 4.18×10^9 J is released during detonation of 1 ton of TNT. How many nuclear fissions occurred at Hiroshima, and what was the total decrease in mass?

Solution: Energy released in detonation of 1 ton of TNT = 4.18×10^9 J
Energy released in detonation of 20,000 tons of TNT

$$= 4.18 \times 10^9 \times 20{,}000 \text{ J}$$

$$= 8.36 \times 10^{13} \text{ J}$$

Energy released in fission of 1 nucleus of ^{235}U = 200 MeV

$$= 200 \times 1.603 \times 10^{-13} \text{ J}$$

$$= 3.206 \times 10^{-11} \text{ J}$$

3.206×10^{-11} J of energy requires 1 fission

8.36×10^{13} J of energy will require $\frac{8.36 \times 10^{13}}{3.206 \times 10^{-11}} = 2.61 \times 10^{24}$ fissions

Total energy released in 2.61×10^{24} fissions = $2.61 \times 10^{24} \times 200$ MeV

$$= 5.22 \times 10^{26} \text{ MeV} = 8.362 \times 10^{13} \text{J}$$

Using the relation $E = mc^2$, we get

$$m = \frac{E}{c^2} = \frac{8.362 \times 10^{13}}{(3 \times 10^8)^2} = 9.29 \times 10^{-4} \text{ kg} = 0.929 \text{ g}$$

= 929 mg

Unsolved Problems

1. Fission of 1 atom of ^{235}U produces about 200 MeV of energy. Estimate the fission rate required to produce 2 watts of energy per second. [**Ans.** 6.25 □ 10^{10} fissions/s].
2. Calculate the energy released if 0.2 kg of ^{235}U undergoes complete fission. [**Ans.** 3.9 □ 10^{12} calories]
3. If each fission of ^{235}U releases 200 MeV, how many fissions occur per second to yield a power level of 2.0 GJ? How much mass is lost per day by the nuclear reactor operated at this power level? [**Ans.** 6.25 □ 10^{19} fissions/s, 1.92 g/day]
4. The fissioning of ^{235}U atom yields 200 MeV of energy. How many atoms of ^{235}U are required to provide an amount of energy equal to that required to lift a housefly by 10 cm? Assume that the mass of the average fly is 5.0 mg. [**Ans.** 1.53 □ 10^{5}]
5. Calculate the energy released in

   (i) Spontaneous fission of ^{235}U to ^{147}La and ^{88}Br.

   (ii) Thermal neutron induced fission of ^{235}U to ^{147}La and ^{89}Br.

   Given:

   $M(^{235}U) = 235.043922$ amu

   $M(^{147}La) = 146.922780$ amu

   $M(^{88}Br) = 87.924070$ amu

   $M(^{89}Br) = 88.926400$ amu [**Ans.** (i) 178.5 MeV, (ii) 184.4 MeV]
6. Calculate the energy released in the spontaneous fission of $^{238}_{92}U$ into two nuclei, each $^{119}_{46}Pd$.

Given:

$M(^{238}U) = 238.0507835$ amu

$M(^{119}Pd) = 118.922680$ amu [**Ans.** 190.5 MeV]

7. Calculate the energy released in the fission of $^{235}_{92}U$ assisted by thermal neutrons into two nuclei, each $^{118}_{46}Pd$. Given:

$M(^{235}U) = 235.043922$ amu

$M(^{118}Pd) = 117.91890$ amu [**Ans.** 200.1 MeV]

8. Upon combustion in air, methane gas undergoes the following chemical reaction:

$$CH_4 + 2O_2 \square CO_2 + 2H_2O$$

During this process about 9 eV/methane molecule energy is released. Find the relative energy released per unit mass for nuclear fission as against this chemical reaction. [**Ans.** $7.6 \square 10^6$]

9. Assume that $^{235}_{92}U$ undergoes fission by a thermal neutron producing $^{97}_{40}Zr$, ^{134}Te and some neutrons. (a) Determine the atomic number of Te from the above data. (b) How many neutrons are released? [**Ans.** 52, 5]

10. $^{235}_{92}U$ undergoes photo fission by *g*-rays of energy of 10 MeV. If the fission products are $^{90}_{36}Kr$, $^{142}_{56}Ba$ and few neutrons, find the number of neutrons emitted in this reaction and write a complete equation for this process. [**Ans.** 3]

11. $^{235}_{92}U$ undergoes photo fission by *g*-rays of energy of 10 MeV. If the fission products are $^{90}_{36}Kr$, $^{142}_{56}Ba$ and three neutrons, calculate the total energy of fission products. Given:

m(U) = 235.0439299 amu

m(Ba) = 141.916457 amu

m(Kr) = 89.919517 amu

m(n) = 1.008665 [**Ans.** 179.5 MeV]

12. A 100 MW reactor consumes half its fuel in three years. How much ^{235}U does it contain? Given that fission of one nuclei yield energy = 200 MeV. [**Ans.** 232.4 kg]

13. Find the ratio of energy released when 1 mg of ^{233}U undergoes fission to the energy released when 1 mg of TNT explodes. Given that

energy released in the explosion of 1 g of TNT is 2.5×10^{22} eV. [**Ans.** 2.06×10^{7}]

14. If an atom bomb of 10 MT explodes, find the total number of fissions that occur and also find the mass which is converted to energy. Assume that 200 MeV is released when a ^{235}U nucleus absorbs a neutron and fissions and that 4.18×10^{9} J is released during detonation of 1 ton of TNT. How many nuclear fissions occurred at Hiroshima, and what was the total decrease in mass? [**Ans.** 1.305×10^{27} fissions, 464 g]

15. ^{236}U undergoes fission into ^{102}Mo and ^{131}Sn isotopes. Complete this reaction. How many neutrons are ejected in this reaction? How many neutrons will be ejected if the fission products are ^{88}Br and ^{140}La isotopes? [**Ans.** $^{236}U + {}^{1}_{0}n \rightarrow {}^{102}_{42}Mo + {}^{131}_{50}Sn + 4{}^{1}_{0}n$, 4 neutrons, $^{236}U + {}^{1}_{0}n \rightarrow {}^{88}_{35}Br + {}^{140}_{57}La + 9{}^{1}_{0}n$, 9 neutrons]

16. If one of the fission fragments from the fission of ^{240}Pu is ^{90}Sr, what is the second fragment, if its mass is 142? How many free neutrons are produced? [**Ans.** $^{240}Pu + {}^{1}_{0}n \rightarrow {}^{90}_{38}Sr + {}^{142}_{56}Ba + 9{}^{1}_{0}n$, 9 neutrons]

17. For the fission reaction

$$^{235}U + n \rightarrow {}^{142}_{55}Cs + {}^{90}_{37}Rb + 4{}^{1}_{0}n$$

estimate the energy released per fission. Given $m_U = 235.0439231$ amu, $m_{Cs} = 141.924292$ amu, $m_{Rb} = 89.914809$ amu and $m_n = 1.008665$ amu. [**Ans.** 166.6 MeV]

18. Calculate the energy released in the symmetric fission of the nuclei with following A and Z values

(i) $A = 238$ and $Z = 92$

(ii) $A = 100$ and $Z = 44$

Given the masses $^{238}_{92}U = 238.050788$ amu, $^{119}_{46}Pd = 118.923100$ amu, $^{100}_{44}Ru = 99.904219$ amu and $^{50}_{22}Ti = 118.923100$ amu. [**Ans.** 190.6 MeV, –5.9 MeV]

19. One gram of ^{235}U upon fission by thermal neutrons releases 2.29×10^{4} kW × h of energy. Find the mass of coal required to produce an

equivalent amount of energy in a thermal power plant. Given that on burning of 1 kg of coal produces 3.5 □ 10^4 kJ of energy. [**Ans.** 2355 kg]

Section 4.8

Solved Problems

1. Calculate the energy liberated if 10 kg of ^{2}H undergoes the following fusion reaction:

$$5\,{}_1^2\mathrm{H} \rightarrow {}_2^3\mathrm{He} + {}_2^4\mathrm{He} + {}_1^2\mathrm{H} + 2\,{}_0^1n + 25\ \mathrm{MeV}$$

Solution: 2 kg of ^{2}H contains = 6.023 □ 10^{26} atoms

10 kg of ^{2}H contains = $\frac{6.023 \times 10^{26}}{2}$ □ 10 = 3.01 □ 10^{27} atoms

Now,

5 atoms of ^{2}H on fusion liberate energy = 25 MeV

3.01 □ 10^{27} atoms on fusion liberate energy = $\frac{25}{5}$ □ 3.01 □ 10^{27} = 1.51 □ 10^{28} MeV

Therefore, 10 kg of ^{2}H in fusion will liberate energy = 1.51 □ 10^{28} MeV = 2.42 □ 10^{15} J.

2. In their old age, heavy stars obtain part of their energy by the reaction:

$${}_2^4\mathrm{He} + {}_6^{12}\mathrm{C} \rightarrow {}_8^{16}\mathrm{O}$$

How much energy does each such event give off?

Solution: Total mass of the reactants = 4.002603 + 12.000000 = 16.002603 amu

Total mass of the product = 15.994915 amu

Difference in the mass = 7.688 □ 10^{-3} amu

Energy produced = 7.688 □ 10^{-3} □ 931.49 = 7.16 MeV.

3. Consider a gas of atoms undergoing fusion. Calculate the temperature required to overcome the Coulomb barrier and the fusion energy

released if the gas consists of
(a) ^{10}B and (b) ^{24}Mg.

Solution:

(a) Let us estimate the height of the Coulomb barrier. It is given by the relation

$$V_{\text{Coul}} = \frac{Z_1 Z_2 e^2}{r}$$

Here, r is the separation between two nuclei at the point of closest approach. It is given by the sum of radii of two ^{10}B nuclei. The radius of each of the nucleus can be estimated using $r = 1.2A^{1/3}$f. Therefore,

$$r = 1.2 \square 10^{1/3} + 1.2 \square 10^{1/3} = 5.17 \text{ f}$$

Coulomb barrier can be written as

$$V_{\text{Coul}} = \frac{e^2}{\hbar c} \times \frac{Z_1 Z_2 \times \hbar c}{r}$$

$$\frac{e^2}{\hbar c} = \frac{1}{137} = a \text{ (fine structure constant)}$$

Substituting various values, we get

$$V_{\text{Coul}} = \frac{1}{137} \times \frac{5 \times 5 \times 197.5 \text{ MeV f}}{5.17 \text{ f}}$$

$$= 6.97 \text{ MeV}$$

$$= 1.12 \times 10^{-12} \text{ J}$$

In order to calculate the temperature required to overcome the Coulomb barrier, we equate this energy to thermal energy as

$$\frac{3}{2} KT = E = V_{\text{Coul}}$$

where K is Boltzmann's constzant and T is absolute temperature.
or

$$= \frac{3}{2} \times 1.38 \times 10^{-23} T \; 1.12 \square 10^{-12}, \text{ which gives}$$

$$T = 5.4 \square 10^{10} \text{ K}$$

Similar calculations are performed for the case of ^{24}Mg fusing with ^{24}Mg as under.

(b) Let us estimate the height of the Coulomb barrier. It is given by the relation

$$V_{\text{Coul}} = \frac{Z_1 Z_2 e^2}{r}$$

Here, r is the separation between two nuclei at the point of closest approach. It is given by the sum of radii of two ^{24}Mg nuclei. The radius of each of the nucleus can be estimated using $r = 1.2A^{1/3}$ f. Therefore,

$$r = 1.2 \square 24^{1/3} + 1.2 \square 24^{1/3} = 6.92 \text{ f}$$

Coulomb barrier can be written as

$$V_{\text{Coul}} = \frac{e^2}{\hbar c} \times \frac{Z_1 Z_2 \times \hbar c}{r}$$

$$\frac{e^2}{\hbar c} = \frac{1}{137} = a \text{ (fine structure constant).}$$

Substituting various values, we get

$$V_{\text{Coul}} = \frac{1}{137} \times \frac{12 \times 12 \times 197.5 \text{ MeV f}}{6.92 \text{ f}}$$
$$= 29.99 \text{ MeV}$$
$$= 4.80 \times 10^{-12} \text{ J}$$

In order to calculate the temperature required to overcome the Coulomb barrier, we equate this energy to thermal energy as

$$\frac{3}{2} KT = E = V_{\text{Coul}},$$

where K is Boltzmann's constant and T is absolute temperature.

or

$\frac{3}{2} \times 1.38 \times 10^{-23} T = 4.80 \square 10^{-12}$, which gives

$T = 23.2 \square 10^{10}$ K

4. Suppose that the sun consists entirely of hydrogen and that the dominant energy-releasing reaction is

$$4{}_1^1\mathrm{H} \rightarrow {}_2^4\mathrm{He} + 2{}_1^0e + 2\nu + g$$

If the total power output of the sun is assumed to remain constant at 3.9 □ 10^{26}W, how long will it take for all of the hydrogen to be burned up? Take the mass of the sun as 2.0 □ 10^{30} kg.

Solution: Total number of hydrogen atoms in the sun = $\dfrac{2.0\times10^{30}\,\mathrm{kg}}{1.67\times10^{-27}\,\mathrm{kg/atom}}$

$$= 1.2 \square 10^{57} \text{ atoms}$$

The given reaction has a mass difference

$$= 4 \square 1.007825 - 4.002603 - 2 \square 0.000549 = 0.027599 \text{ amu}$$

which in terms of energy is 0.027599 □ 931.47 = 25.7 MeV

Since each fusion reaction consumes 4 hydrogen atoms, the total energy available in the sun

$$= \frac{1.2\times10^{57}}{4} \times 25.7 \text{ MeV}$$

$$= 7.71 \square 10^{57} \text{ MeV}$$

$$= 1.24 \square 10^{45} \text{ J}$$

3.9 □ 10^{26} J of energy is released in 1s.

1.24 □ 10^{15} J of energy will be released in $\dfrac{1.24\times10^{45}}{3.9\times10^{26}} = 3.18\times10^{18}$ s.

= 1.008 □ 10^{11} years

Thus, this gives an estimated lifetime of about 100.8 billion years.

5. Calculate the mass defect and *Q*-values for the fusion reactions

$$d + d \square {}^3\mathrm{He} + n$$

$$d + d \square {}^3\mathrm{H} + p$$

Assuming these occur with the deuterons at rest, find the kinetic energies of the outgoing particles in each case. Given m_p = 1.007825 amu, m_n

= 1.008665 amu, $m(^2\text{H})$ =

2.014102 amu, $m(^3\text{H})$ = 3.016049 amu and $m(^3\text{He})$ = 3.016029 amu.

Solution: We have

$$d + d \rightarrow {}^3\text{He} + n$$

Mass defect is given by the relation

$$\text{Mass defect} = 2 \times m_d - m(^3\text{He}) - m_n$$

Substituting various given masses, we get

$$\text{Mass defect} = 2 \times 2.014102 - 3.016029 - 1.008665$$
$$= 0.00351 \text{ amu}$$

and

$$Q\text{-value} = \text{mass defect (in amu)} \times 931.47 \text{ MeV}$$
$$= 0.00351 \times 931.47 \text{ MeV}$$
$$= 3.27 \text{ MeV}$$

$$d + d \rightarrow {}^3\text{H} + p$$

Mass defect is given by the relation

$$\text{Mass defect} = 2 \times m_d - m(^3\text{H}) - m_p$$

Substituting various given masses, we get

$$\text{Mass defect} = 2 \times 2.014102 - 3.016049 - 1.007825$$
$$= 0.00433 \text{ amu}$$

and

$$Q\text{-value} = \text{mass defect (in amu)} \times 931.47 \text{ MeV}$$
$$= 0.00433 \times 931.47 \text{ MeV}$$
$$= 4.03 \text{ Mev}$$

Assuming the initial state deuterons are essentially at rest then the final state kinetic energy is equal to Q. By applying the conservation of momentum it can be seen that the share of the kinetic energy that each particle has is inversely proportional to its mass. Thus, for these reactions, the heavier particle takes one quarter while the lighter particle takes three quarters of the total kinetic energy (Q).

Unsolved Problems

1. Calculate the energy librated in the following reaction:

$p + {}^{11}\mathrm{B} \rightarrow 3\,{}^{4}_{2}\mathrm{He}$

Given:

$M(p) = 1.007825$ amu

$M({}^{4}\mathrm{He}) = 4.002603$ amu

$M({}^{11}\mathrm{B}) = 11.009305$ amu [**Ans.** 8.68 MeV]

2. ${}^{2}\mathrm{H}$ and ${}^{3}\mathrm{H}$ undergoes the following fusion reaction:

$$ {}^{2}\mathrm{H} + {}^{3}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}n + 17.6 \text{ MeV} $$

Calculate the amounts of ${}^{2}\mathrm{H}$ and ${}^{3}\mathrm{H}$ required to produce 10^{16} J of energy. [**Ans.** 11.8 kg, 17.7 kg]

3. Upon combustion in air, methane gas undergoes the following chemical reaction:

$$ \mathrm{CH_4} + 2\mathrm{O_2} \rightarrow \mathrm{CO_2} + 2\mathrm{H_2O} $$

During this process about 9 eV/methane molecule energy is released. In sun, $4\,{}^{1}_{1}\mathrm{H}$ atoms undergoes fusion as under:

$$ 4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{0}_{+1}e + 2\nu_e + 26.73 \text{ MeV} $$

Calculate the relative energy released per unit mass for nuclear fusion as against this chemical reaction. [**Ans.** 5.9×10^{7}]

4. The energy released during the explosion of hydrogen bomb has been estimated as equivalent to that released by 50 mega tons of TNT. Assume that the following reaction takes place in the bomb

$$ 4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{0}_{1}e + 2\nu $$

Find the weight of hydrogen required to produce the required power output. Also, find the weight of hydrogen converted to energy. Given that 4.18×10^{9} J is released during detonation of 1 ton of TNT, $m_{\mathrm{p}} = 1.007276$ amu, $m_{\mathrm{He}} = 4.0015$ amu and $m_2 = 0.00054858$ amu. [**Ans.** 353 kg, 2.32 kg]

5. In 2006, total world wide energy consumption was about 6 □ 10^{20} J per year. Suppose this entire energy is derived from the reaction

$$3\,{}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{1}\mathrm{H} + {}^{1}_{0}n$$

How many deuteron atoms would be required to generate this energy? What would be the weight of these atoms? Given m_d = 2.0141022 amu, m_a = 4.002603 amu, m_n = 1.00866501 amu and m_p = 1.00727647 amu. [**Ans.** 5.07 □ 10^{32}, 1684 tons]

6. In a fusion reaction six deuterons fuse to give 2 ^{4}He nuclei, two protons and two neutrons. Total energy released in this process is 43.2 MeV. Calculate the energy released by fusion of 2 g of deuterons. [**Ans.** 4.34 □ 10^{24} MeV]

REVIEW QUESTIONS

Short Answer Type

1. Define nuclear reaction cross-section. What are its units and dimensional formula?
2. What is nuclear reaction cross-section?
3. What is differential cross-section of a nuclear reaction?
4. What are stripping and pickup reactions?
5. What is spalliation nuclear reaction?
6. What is nuclear transmutation?
7. Name and explain the physical quantities that are not conserved in a nuclear reaction.
8. What are the quantities which remain conserved in a nuclear reaction?
9. How do you support the formation of compound nucleus in a nuclear reaction?
10. What is half-thickness of a material?
11. Differentiate between nuclear decay process and a nuclear reaction.
12. For a nuclear reaction ${}^{24}_{12}\mathrm{Mg}(a, n){}^{27}_{14}\mathrm{Si}$, how do you think, we can calculate the Q-value from the masses?
13. What does the X in the following reaction represent? Also write about the compound nucleus formed.

$^{14}_{7}\mathrm{N}(X, p)^{17}_{8}\mathrm{O}$

14. What is meant by Q-value of a nuclear reaction?
15. Can a nuclear reaction be induced by photons? Explain giving an example.
16. Name various types of nuclear reactions.
17. Define the differential and total scattering cross-sections.

Long Answer Type

1. What is kinematics of nuclear reaction? What is the Q-value and its significance?
2. What is the Q-value of a nuclear reaction? Derive an expression for Q-value of a nuclear reaction in terms of masses and kinetic energies of incident particle, product particle and nuclei.
3. What is a compound nucleus? Who proposed it? What is its significance?
4. Define nuclear cross-section. Derive an expression for number of surviving particles, if a number of particles is incident on a slab of certain area, thickness and number density.
5. What is a nuclear reaction? Define its cross-section and Q-value. Derive an expression for the Q-value of the reaction $X(a, b)Y$ in terms of the kinetic energy of particles.
6. Write notes on:
   (i) Nuclear reaction cross-section.
   (ii) Compound nucleus and assumptions.
7. Define the threshold energy for a nuclear reaction and derive an expression for it for the reaction $X(x, y)Y$ in terms of the Q-value and masses of the participating nuclei. Write the condition for exothermic and endoergic nuclear reactions.
8. Explain the terms nuclear reaction cross-section and differential cross-section. Derive an expression for nuclear reaction cross-section. Has the nuclear reaction cross-section any relation to the target area?
9. What is a nuclear reaction? Discuss various conservation laws in nuclear reactions with illustrative examples.
10. Define and calculate the threshold energy of an endoergic nuclear reaction that proceeds through formation of a compound nucleus.

11. What is a nuclear reaction? Give two examples of nuclear reactions produced in each case by a-particle, proton and neutron.

12. Discuss possible reactions that may occur when high energy charged particle approaches nucleus.

13. What are nuclear reactions? State conservation laws obeyed in such reactions. What is meant by Q-value of a reaction?

14. Discuss by giving examples the different types of nuclear reactions.

15. Explain compound nucleus theory of nuclear reactions. What do you mean by cross-section of a nuclear reaction?

5.5 INTERACTION OF GAMMA RAYS WITH MATTER

If gamma ray passes through an absorber of thickness Δx, its intensity I changes. The change in intensity ΔI, is proportional to the intensity I of the g-rays falling on the absorber and thickness of the absorber. The change in intensity is given by

$$\Delta I \mu - I \Delta x$$

or

$$\Delta I = - mI \Delta x \qquad (5.9)$$

Negative sign indicates that as x increases, I decreases. Here m is constant of proportionality and is known as *absorption coefficient* or *attenuation coefficient*. For a given absorber, m depends upon the energy of gamma rays. Equation (5.9) can be written as

$$\frac{\Delta I}{I} = -m \Delta x$$

which on integration gives

$$\ln(I) = -m x + c \qquad (5.10)$$

when $x = 0$, $I = I_0$. Here I_0 is the initial intensity of g-rays falling on the absorber.

From Eq. (5.10)

$$c = \ln(I_0)$$

Substituting for c in Eq. (5.10), we get

$$\ln \left(\frac{I}{I_0}\right) = -m x$$

This equation can also be written as

$$I = I_0 e^{-m x} \qquad (5.11)$$

Absorption coefficient m is also called *linear absorption coefficient*. Sometimes we also use the term mass absorption coefficient m_m, which is defined as

$m_m = \frac{m}{r}$ where r is the density of the absorber.

Since in Eq. (5.11) mx must be a dimensionless quantity, therefore, if x is expressed in metres, then m will have the dimensions of metre^{-1}.

From Eq. (5.11), it is clear that intensity of *g*-rays decreases exponentially and never becomes zero. Interaction of *g*-rays is markedly different from that of charged particles. *g*-rays have greater penetrating power and different absorption laws.

As stated earlier, charged particles suffer a large number of collisions before finally stopped by the medium. In *g*-rays and thin absorber, each photon is removed individually in a single event. The number of photons removed in thickness Δx is proportional to Δx and also proportional to the number of photons reaching Δx. This dependence leads to Eq. (5.11).

In Eq. (5.11), if $x = \frac{1}{m}$, then we get

$$I = I_0 e^{-1}$$

or

$$\boxed{I = \frac{I_0}{e}}$$

Therefore, linear attenuation coefficient is also defined as reciprocal of that thickness of the material which reduces the intensity of the incident *g*-rays to $\frac{1}{e}$ of its original value.

5.5.1 Radiation Length

Reciprocal of linear attenuation coefficient is known as *radiation length*. It is also known as *absorption length*. The radiation length is also defined as the thickness of the material which reduces the intensity of the incident *g*-rays to $\frac{1}{e}$ of its original value. If the units of attenuation coefficient m are metres^{-1}, then the units of radiation length are metres.

5.5.2 Half-thickness

Half-thickness of an absorber is that thickness which reduces the intensity of *g*-ray beam by
one-half of its original value.

From Eq. (5.11)

$$I = \frac{I_0}{2}$$

$$\therefore \quad \frac{I_0}{2} = I_0 e^{-\mu x}$$

or

$$e^{\mu x} = 2$$

Taking $\log_e$ on both sides, we get

$$m\, x_{1/2} = \ln 2$$

where $x_{1/2}$ is half-thickness, i.e. the thickness which reduces the intensity of *g*-rays by one-half of its original value. This can also be written as

$$\boxed{m = \frac{0.6931}{x_{1/2}}} \qquad (5.12)$$

5.5.3 Experimental Determination of Attenuation Coefficient

Experimentally, linear attenuation coefficient is determined using Eq. (5.11), which can be written as

$$\frac{I}{I_0} = e^{-} m\, x$$

or

$$-m\, x = \ln \left(\frac{I}{I_0}\right)$$

This relation is a straight line between thickness x and $\ln \left(\frac{I}{I_0}\right)$. Slope of this straight line gives m.

The apparatus used to determine the value of attenuation coefficient is shown in Figure 5.3. A radioactive source say ^{137}Cs, (*S*) which emits *g*-rays of energy 662 keV, is enclosed in a thick block of lead with one very fine opening on one side as shown in Figure 5.3. From this arrangement we get a very fine beam of *g*-rays, which passes through a lead collimator A. This fine beam of *g*-rays passes through the absorber *B* for which attenuation coefficient is to be measured. Any scattered photons from the absorber *B* are again stopped by another lead collimator *C*. Only beam of *g*-rays attenuated

and transmitted through the absorber reaches the heavily shielded *g*-ray detector *D*.

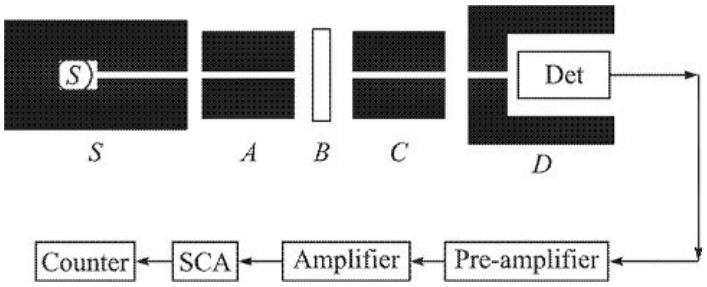
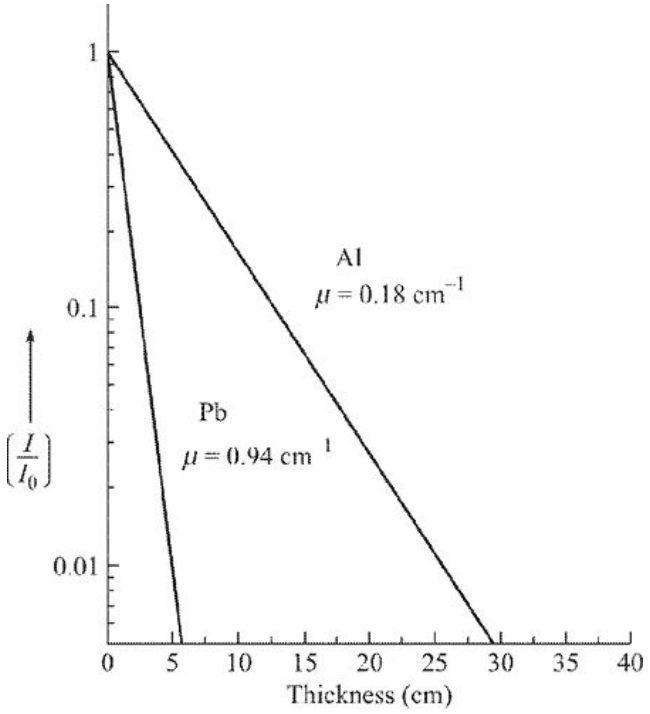


Figure 5.3 Block diagram of the apparatus used for determination of attenuation coefficients.

The experiment is first performed without putting any absorber at position *B*. The counts recorded in the *g*-ray spectrometer give I_0. Then by putting absorbers of different thicknesses at position *B*, corresponding counts *I* are recorded. A plot between ln $\left(\frac{I}{I_0}\right)$ versus thickness of the absorber *x* is shown in Figure 5.4 for Al and Pb absorbers. The slope of this line gives the value of *m*, which in this case is 0.18 cm^{-1} for Al and 0.94 cm^{-1} for Pb.



space and forms e^- and e^+ pair. The vector diagram of momenta involved when photon is converted into electron–positron pair is shown in Figure 5.12. Applying energy conservation, we get

$$hn = 2mc^2 \qquad (5.30)$$

where m is the moving mass of moving electron or positron.

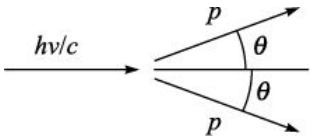


Figure 5.12 Momentum conservation in pair production.

The angles of emission of e^- and e^+ are equal due to conservation of linear momentum along perpendicular direction.

Applying momentum conservation in the direction of incident photon, we get

$$\frac{h\nu}{c} = 2p \cos \theta \qquad (5.31)$$

or

$$h\nu = 2pc \cos \theta$$

and

$$p = mv \qquad (5.32)$$

where n is the velocity of both electron and positron. This equation can also be written as

$hn = 2mc^2 \frac{v}{c} \cos q$ (5.33)

As $\frac{v}{c} < 1$ and cos q □□1, therefore,

$$hn \square\square 2mc^2 \qquad (5.34)$$

This contradicts with the law of energy conservation ($hn = 2mc^2$). Therefore, it is not possible for pair production to take place in vacuum as it violates energy, momentum conservation laws. Presence of third body—the nucleus is a must which carries away part of the momentum of the incident photon, thereby satisfying both the laws.

NUMERICAL PROBLEMS

Section 5.2

Solved Problem

1. A proton of energy 10 MeV is moving in Pb. How much energy does it lose in a collision with an electron? How many collisions are required to completely stop it?

Solution: Energy lost in a single collision with an electron = $\frac{4m}{M}E$

$$= \frac{4 \times 0.511 \times 10}{938}$$

$$= 0.02179 \text{ MeV}$$

$$= 21.79 \text{ keV}$$

Total number of collisions required to completely stop proton = $\frac{10,000 \text{ keV}}{21.79 \text{ keV/collision}}$

$$= 459 \text{ collisions}$$

2. The scattering of an energetic charged particle in matter is mostly due to interactions with (a) electron (b) nuclei (c) quarks

Solution: The incident charged particle undergoes Coulomb interaction with both electrons and nuclei. The collisions with electrons are numerous, but the momentum transfer is small and therefore suffers very small scattering. Only collisions with the nuclei result in appreciable scattering of the incident particle. Therefore, scattering of energetic charged particles is due to collisions with nuclei.

Unsolved Problems

1. An oxygen ion of energy 10 MeV is moving in Pb. What is the energy lost by oxygen ion in a single collision with electron? How many such collisions are required to stop the oxygen ion? [**Ans.** 1.36 keV, 7342 collisions]
2. A proton of energy (i) 2 MeV, (ii) 5 MeV is moving in Al. How much energy does it lose in a collision with an electron? [**Ans.** 4.36 keV, 10.90 keV]

Section 5.3

Solved Problem

1. Compare the radiation losses with the ionization losses for 0.9756, 9.756 and
97.56 MeV *b*-particles in Pb.

Solution: We have from Eq. (5.7)

$$\frac{dE/dX)_{\text{radiation}}}{dE/dX)_{\text{ionization}}} = \frac{Z(T + m_0c^2)}{m_0c^2 \times 1600}$$

Therefore, for *b*-particles of energy 0.9756 MeV

$$\frac{dE/dX)_{\text{radiation}}}{dE/dX)_{\text{ionization}}} = \frac{82(0.9756 + 0.511)}{0.511 \times 1600} = 0.15$$

Similarly, for *b*-particles of energy 9.756 MeV

$$\frac{dE/dX)_{\text{radiation}}}{dE/dX)_{\text{ionization}}} = \frac{82(9.756 + 0.511)}{0.511 \times 1600} = 1.03$$

and for *b*-particles of energy 97.56 MeV

$$\frac{dE/dX)_{\text{radiation}}}{dE/dX)_{\text{ionization}}} = \frac{82(9.756 + 0.511)}{0.511 \times 1600} = 9.84$$

Unsolved Problems

1. Compare the radiation losses with the ionization losses for 20.0 MeV *b*-particles in Al and Pb. What conclusion you draw from the results? [**Ans.** 0.33, 2.06]
2. Calculate the *b*-energy for which radiation and ionization losses are equal in Pb. Repeat the calculations for Al. [**Ans.** 10.48 MeV, 63.40 MeV]
3. For Z = 10, 20, 30, 40, 50, 60, 70, 80 and 90, what are the values of *b*-energies for which radiation and ionization losses are equal? Plot E versus Z for this case. [**Ans.** 81.3, 40.4, 26.7, 19.9, 15.8, 13.1, 11.1, 9.7, 8.6]

Section 5.5

Solved Problems

1. In an absorption experiment with 1.14 MeV *g*-radiation from ^{65}Zn, it is found that 20 cm of Al reduced the beam intensity to 3%. Calculate the half-value thickness of Al for this radiation.

Solution:

Given:

$$\frac{I}{I_0} = \frac{3}{100}$$

Thickness of Al = 20 cm = 0.2 m. We have from Eq. (5.11)

$$I = I_0 e^{-\mu x}$$

or

$$\frac{I}{I_0} = e^{-\mu x}$$

or

$$\frac{3}{100} = e^{-0.2 \times \mu}$$

or

$$\frac{100}{3} = e^{0.2 \times \mu}$$

Taking $\log_e$ on both sides, we get

$$3.50656 = 0.2m$$

which gives

$$m = 17.533 \text{ m}^{-1}$$

Half-value thickness is given by

$$x_{1/2} = \frac{0.6931}{\mu}$$
$$= \frac{0.6931}{17.533}$$
$$= 0.0395 \text{ m}$$

2. Linear absorption coefficient of Pb for 1 MeV *g*-rays is 0.75 cm^{-1}. Calculate thickness of lead required to reduce the intensity of *g*-rays to $\frac{1}{100}$ times.

Solution:

Given:

$\frac{I}{I_0} = \frac{1}{100}$ and $m = 0.75\ \text{cm}^{-1}$

We have from Eq. (5.11)

$$I = I_0 e^{-\mu x}$$

or

$$\frac{I}{I_0} = e^{-\mu x}$$

$$\Rightarrow \quad \frac{I_0}{I} = e^{\mu x}$$

or

$$\frac{100}{1} = e^{0.75 \times \mu}$$

Taking $\log_e$ on both sides, we get

$$4.6052 = 0.75x$$

or $x = 6.579$ cm

3. Half-thickness of an absorber of *g*-rays is 5 mm. What is the percentage loss of intensity of these *g*-rays in travelling 20 mm thickness of the absorber?

Solution:

Given:

$$x_{1/2} = 5 \text{ mm}$$

We have from Eq. (5.12)

$$x_{1/2} = \frac{0.6931}{\mu}$$

or

$$\mu = \frac{0.6931}{5} \text{ mm}^{-1}$$

or

$$\mu = 0.13862 \text{ mm}^{-1}$$

$$I = I_0 e^{-\mu x}$$

or

$$\frac{I}{I_0} = e^{-\mu x}$$

$$= e^{-0.13862 \times 20}$$

$$\frac{I}{I_0} = 0.0625$$

$$I = 0.0625 I_0$$

Thus, the intensity of *g*-rays will reduce to 6.25% of its original value.

4. ^{24}Na emits two *g*-rays of energy 1378 keV and 2754 keV respectively. Calculate the relative intensities of the two *g*-rays after they have passed through 55.0 g/cm^2 of Pb (r = 11.0 g/cm^3). The linear absorption coefficients are 0.5 cm^{-1} for the 2754 keV and 0.6 cm^{-1} for the 1378 keV *g*-rays.

Solution: The thickness of Pb sheet = $\frac{55.0 \text{ g/cm}^2}{11.0 \text{ g/cm}^3} = 5.0 \text{ cm}$

The attenuation of the two *g*-rays is given by

$$\frac{I_1}{I_0} = e^{-\mu_1 x}$$

and

$$\frac{I_2}{I_0} = e^{-\mu_2 x}$$

Therefore, on dividing these two relations, we get

$$\frac{I_1}{I_2} = \frac{e^{-\mu_1 x}}{e^{-\mu_2 x}}$$

$$= e^{(\mu_2 - \mu_1)x}$$

or

$$\frac{I_1}{I_2} = e^{(0.6-0.5)\times 5.0}$$

$$= e^{0.5}$$

$$= 1.65$$

5. Find the half-thickness for *b*-particles from ^{210}Bi in Al. The end point energy of

b-particles from ^{210}Bi is 1.17 MeV. Given absorption coefficient *m* of Al at this energy is 17.8 cm^2/g.

Solution: Half-value thickness is defined as that thickness of absorber which reduces the intensity to one-half of its original value. The relations for calculating half-value thickness given for *g*-rays are also applicable to *b*-particles. Therefore, from Eq. (5.11)

$$\frac{I}{I_0} = \frac{1}{2} = e^{-\mu x_{1/2}}$$

Therefore,

$$x_{1/2} = \frac{\log 2}{\mu}$$

$$= \frac{0.693}{17.8}$$

$$= 0.03893 \frac{\text{g}}{\text{cm}^2}$$

In order to convert this thickness from $\frac{\text{g}}{\text{cm}^2}$ to usual units cm, let us divide it by density of Al (= 2.7 g/cc). Therefore,

$$x_{1/2} = \frac{0.03893 \text{ g/cm}^2}{2.7 \text{ g/cc}}$$

$$= 0.0144 \text{ cm}$$

6. A beam of antineutrinos from a reactor is incident on earth. Find the absorption length, i.e. the distance in which the beam is reduced by a

factor 1/*e* of its original value. Given density of earth = density 5000 kg m^{-3}, cross section 10^{-19} barn per electron. (Ignore interactions with nuclei and assume that *Z*/*A* is approximately a half.)

Solution: As antineutrinos interact with electrons present in the earth, let us first estimate the number of electrons present in 1 m^3 of soil.

The weight of 1 m^3 of earth is equal to its density = 5000 kg.

The number of nucleons (protons + neutrons) present in 1 m^3 of earth is calculated by dividing the weight of the earth (5000 kg) by mass of nucleon 1.674×10^{-27} kg.

Therefore, the number of nucleons present in 1 m^3 of earth = $\frac{5000}{1.674 \times 10^{-27}}$.

The number of protons present $= \frac{1}{2}$ of total nucleons present

$$= \frac{1}{2} \times \frac{5000}{1.674 \times 10^{-27}}$$

This number must be equal to the number *n* of electrons present as the earth is electrically neutral.

Therefore,

$$n = \frac{1}{2} \times \frac{5000}{1.674 \times 10^{-27}}$$

Now we have

$$x = l = \frac{1}{n\sigma}$$

Substituting various values, we get

$$x = \frac{1}{\frac{5000}{2 \times 1.674 \times 10^{-27}} \times 10^{-19} \times 10^{-28}} \text{ m}$$

$$= 6.7 \times 10^{16} \text{ m} = 6.7 \times 10^{13} \text{ km}$$

It is worth mentioning here that this thickness is approximately equal to the distance of nearest star (Alpha Centauri) from the earth. This distance is approximately equal to about 4.35 light years.

7. The thickness of Cd sheet required to reduce the intensity of slow neutrons to 2% of its original value is 0.03 cm. Find the neutron absorption cross section of Cd. Given density of Cd = 8.7 g/cm^3, its atomic weight = 112.

Solution: The number of Cd atoms per unit volume is

$$n = \frac{\rho}{A} N_{av}$$

$$= \frac{8.7}{112} \times 6.023 \times 10^{23}$$

$$= 4.679 \times 10^{22} \text{ cm}^{-3}$$

The required thickness t of Cd foil is given by

$$I = I_0 \, e^{-n\sigma t}$$

or

$$t = \frac{1}{n\sigma} \ln \frac{I_0}{I}$$

or

$$\sigma = \frac{1}{nt} \ln \frac{I_0}{I}$$

$$\sigma = \frac{1}{4.7 \times 10^{22} \times 0.03} \ln \frac{1}{0.02}$$

$$= 2.7745 \times 10^{-21} \text{ cm}^2$$

$$= 2774.5 \text{ barn}$$

Unsolved Problems

1. How much Pb is required to reduce the intensity of 500 keV g-ray to 50% of its initial intensity? Given: m for Pb = 164 m^{-1} [**Ans.** 0.0042 m]
2. How much air column is required to reduce the intensity of 500 keV g-ray to 1% of its initial intensity? Given: m for air = 0.0112 m^{-1}. [**Ans.** 411.2 m]
3. Dentists commonly use 50 kV X-ray machines generating X-rays of 50 keV. Estimate the thickness of Pb sheet that absorbs 99.99% of 50 keV X-rays. Given linear attenuation coefficient for X-rays of 50 keV in Pb is 93.2 cm^{-1}. [**Ans.** 0.99 mm]